

MPLS

Multi-protocol label switching



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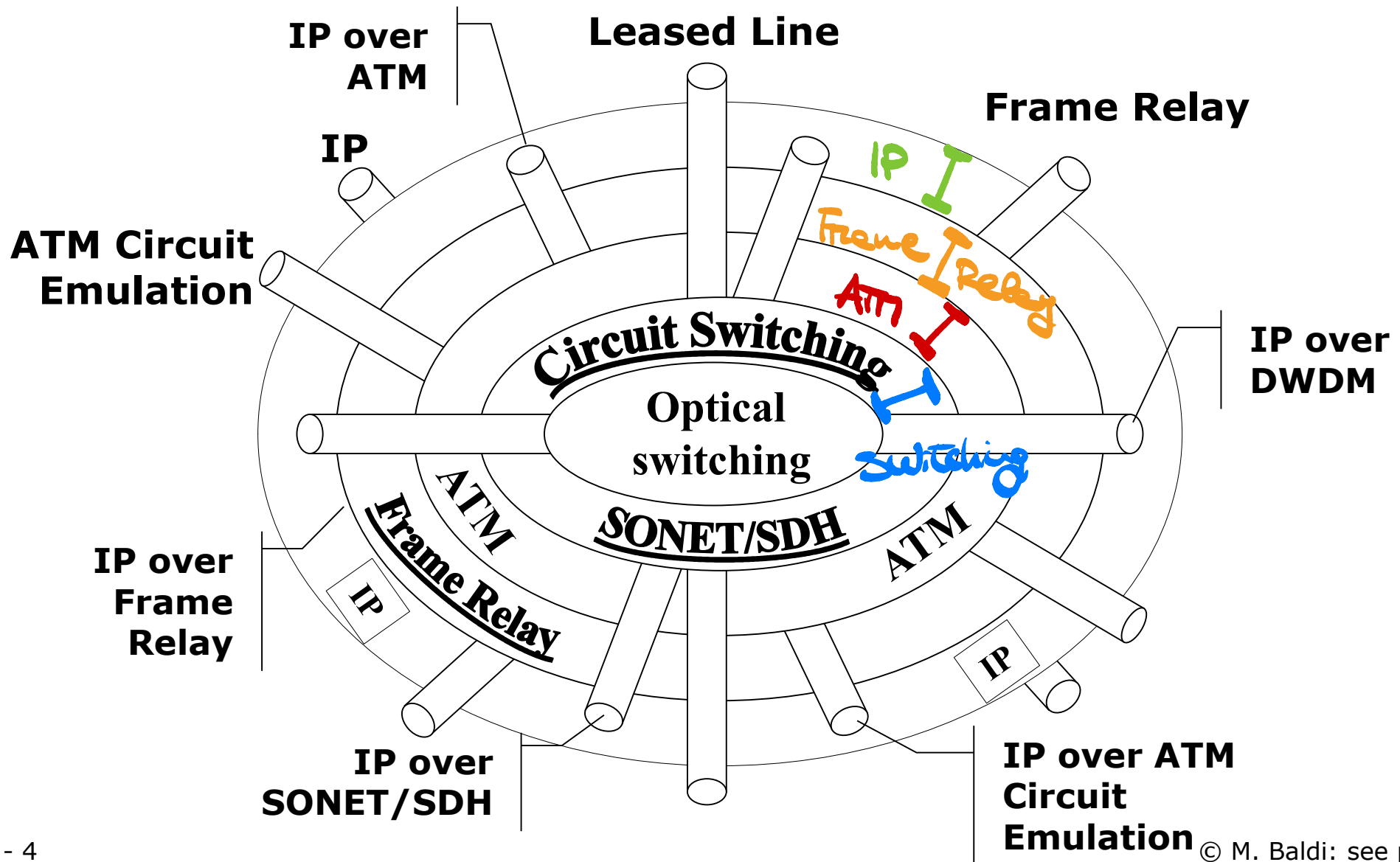
From MPLS Forum Documents

It allows to control the quality of the service

Multi-Protocol Label Switching

MPLS is the enabling technology for the New Broadband (IP) Public Network

The “onion” that makes telecos “cry”



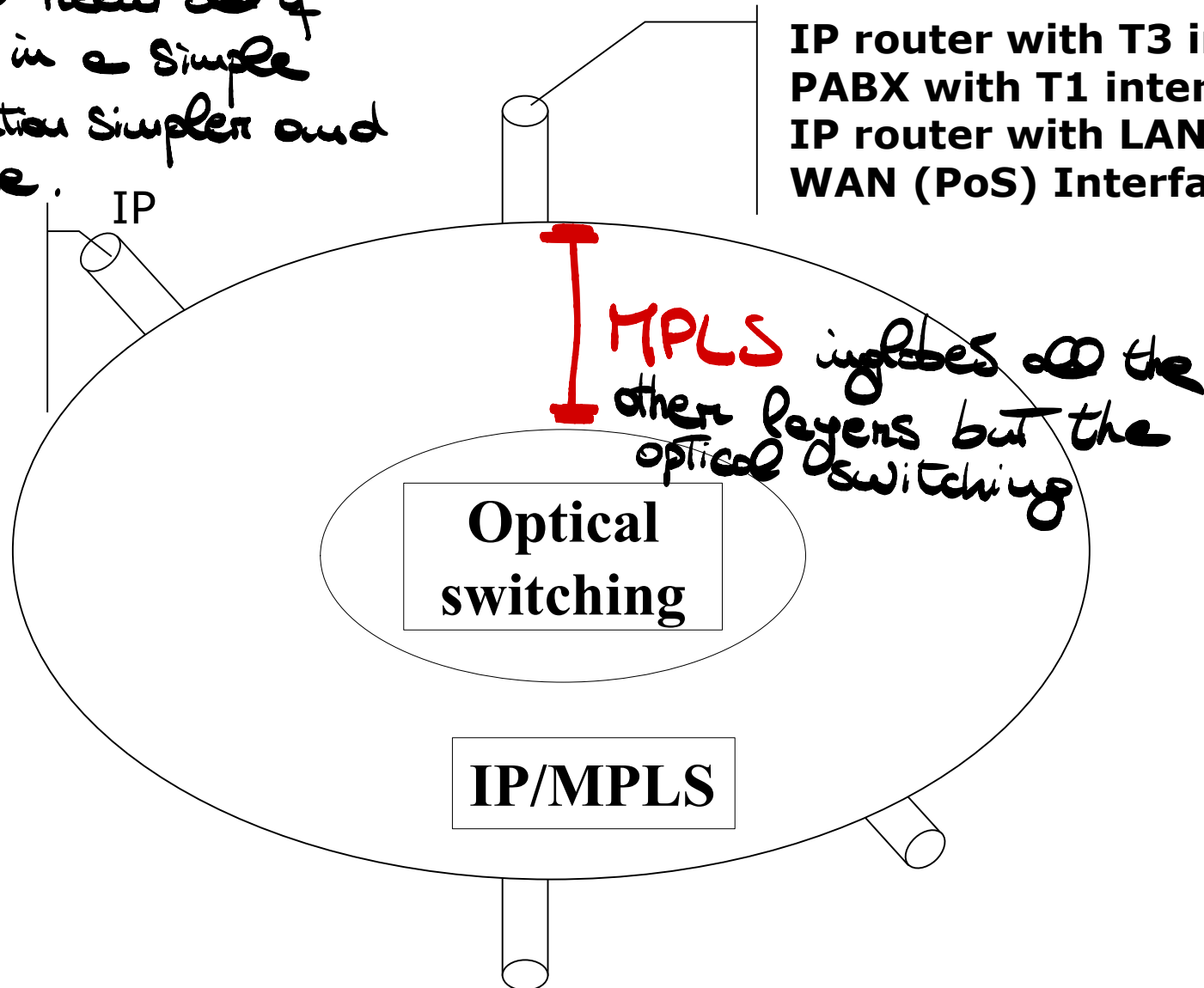
The future: WDM, IP, MPLS

MPLS allows to read all of those layers in a simple way. One solution simpler and less expensive.

IP router with LAN or WAN interface

IP

IP router with T3 interface
PABX with T1 interface
IP router with LAN (GE) or WAN (PoS) Interface



The idea



**Forwarding packets
according to a label**

**Instead of the IP
destination address**

Why?

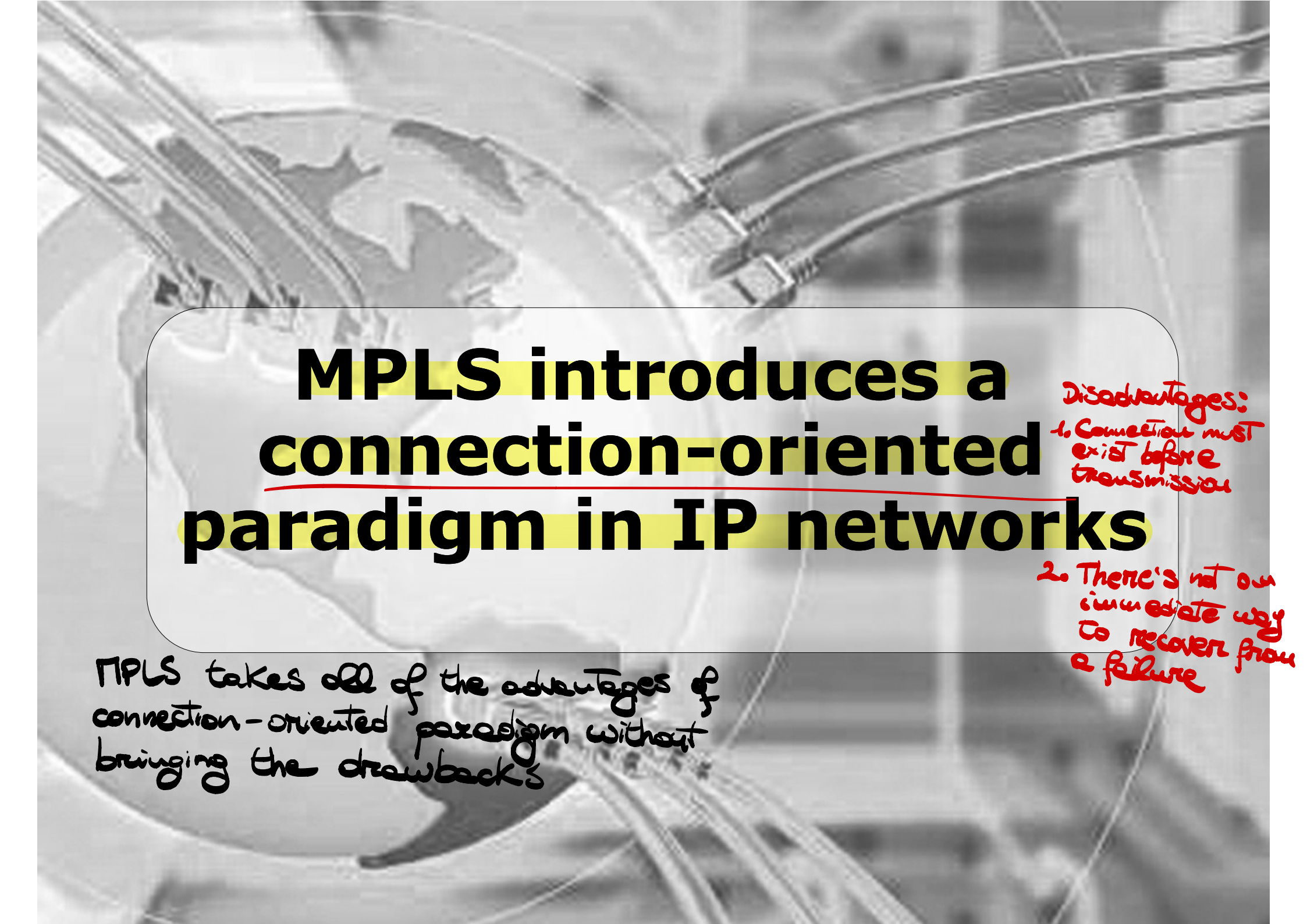
Faster lookup

→ The label may be used as an index

→ Instead of longest prefix matching

made by TCAM (extremely expensive and uses a lot of power)

Traffic engineering



MPLS introduces a connection-oriented paradigm in IP networks

Disadvantages:

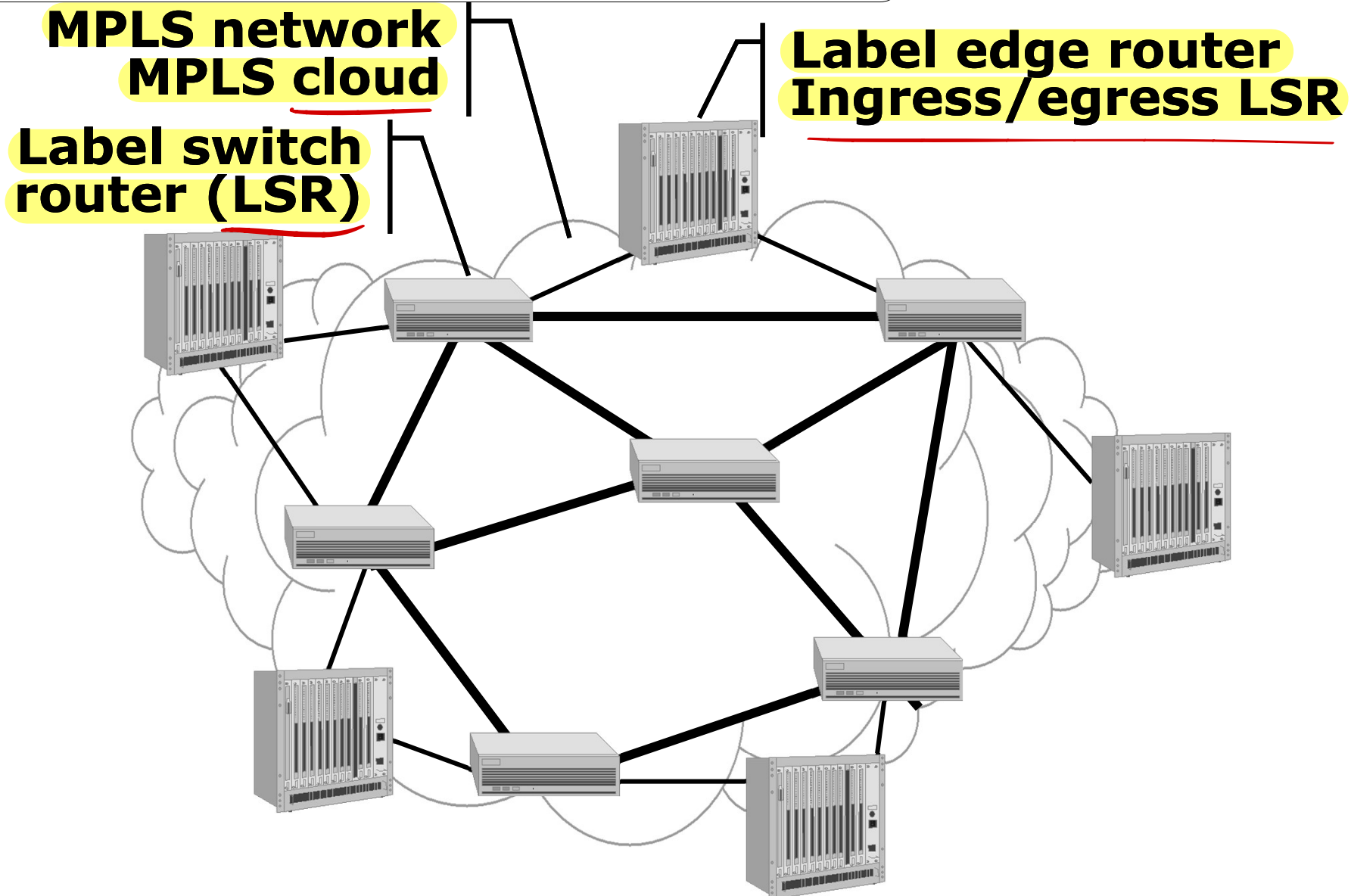
1. Connection must exist before transmission

2. There's not an immediate way to recover from a failure

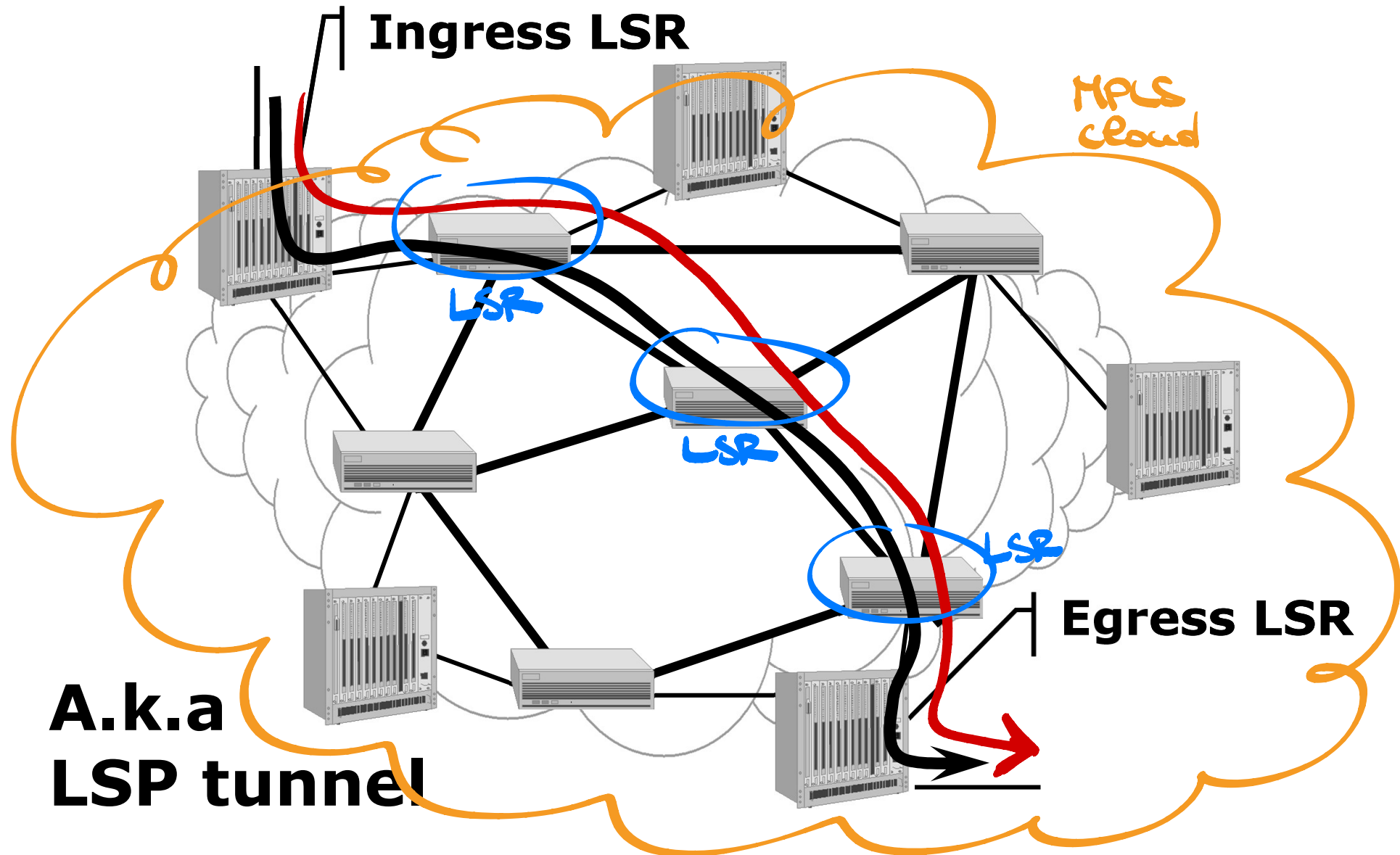
MPLS takes all of the advantages of connection-oriented paradigm without bringing the drawbacks

Network architecture

We don't know about MPLS because it runs only in the backbone of the network, not on end devices



Label Switched Path (LSP)



Label switching

Label
Packet

A	Y	
D	Y	
E	R	

In	Out
	Z
	Z
	R

Forwarding table

In	Out
	R
	W

In	Out
	T
	Q

In	Out
	U
	V

Here the label is removed and we get the normal packet



If I change label at every hop, I don't need to have a global knowledge of the labels used in every single route, but only the ones used in that route (at most the number of hop in that route)

MPLS key elements

→ MPLS "header" Label

→ Contains the label

→ Protocols for label distribution

Routers agree on which labels to use

→ Signaling

→ Enhanced routing protocols

→ Constraints in choosing paths

It uses the same IP routing protocols but they're enhanced to include additional information in order to satisfy certain constraints

Solution that satisfied the needs
of information technology

MPLS history

→ **Tag Switching**

→ **IP over ATM** Packet technology to design Broadband Public Networks

Compared to ATM

→ **No problems with resolution of addresses**

→ **Simpler signaling**

→ **Only one control plan**

→ **ATM with IP**

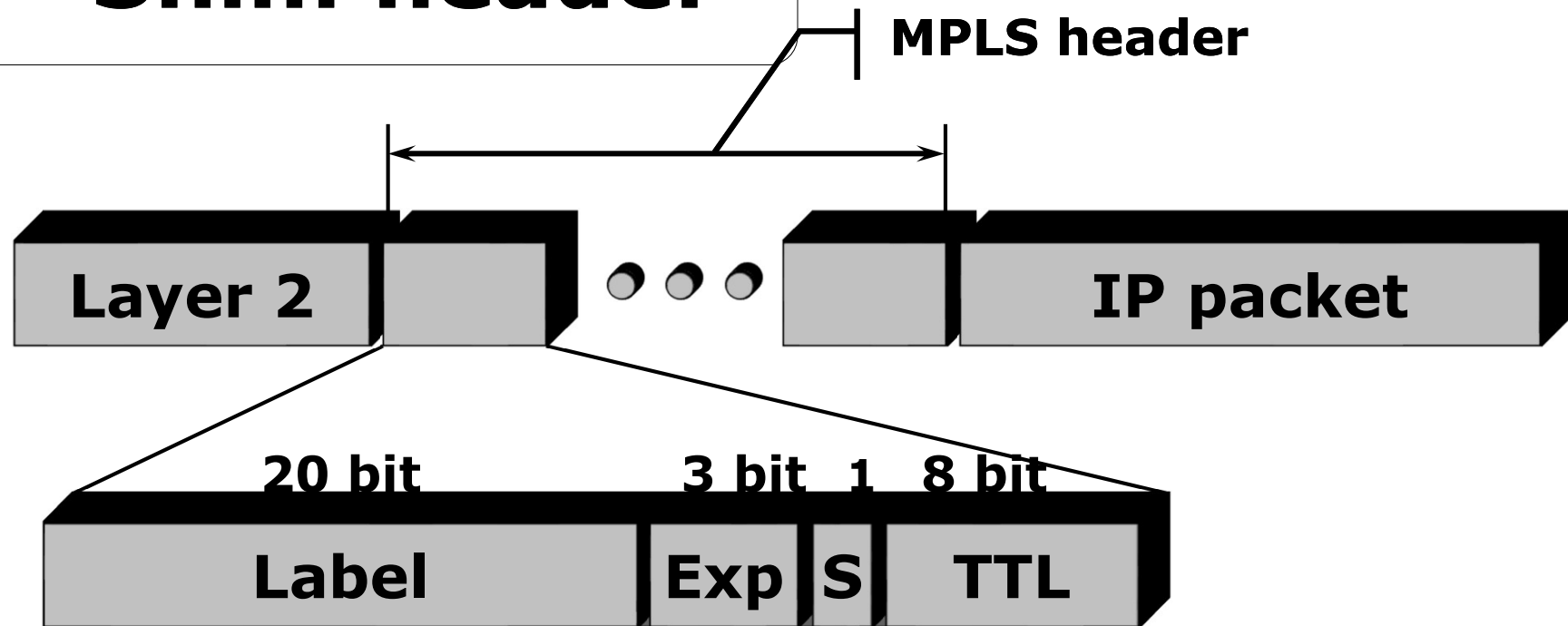
→ **Reusing ATM switching hardware**

MPLS history

- **MP λ S (Multi-Protocol Lambda Switching)**
- **G-MPLS (Generalized MPLS)**
 - **Packet switching**
 - **Cell switching**
 - **Circuit switching (SONET/SDH)**
 - **Lambda switching**
 - **Anything switching**
- Unifying control plane**

MPLS “header”

Shim header



Exp: Experimental bits (CoS)
S: Bottom of stack
TTL: Time to live

ATM and frame relay

Connection oriented layer 2 protocols

MPLS labels in layer 2 header

- **VCI/VPI (ATM)**
- **DLCI (Frame relay)**

LSP setup: label and path selection

A glimpse into the control plane

The header is managed in
the data plane

Forwarding equivalence class (FEC)

Packets that

**Are treated the same way
by each LSR**

**Follow the same path
in MPLS network**

Receive the same label LSP

Three Key Actions

- **Label binding**
- **Label mapping**
- **Label distribution**

Taken by LSRs

Label Binding

An LSR determines the label that should be prepended to packets belonging to a given FEC

→ **Downstream binding** *router* *Only one of the two routers will decide which label use*

→ **LSR on receiving end of a link**

→ **Packets of the FEC shall be received with chosen label**

→ **Upstream node to be notified**

→ **Unsolicited**

→ **On-demand**

Label Distribution

Notification of the label chosen for a given FEC

→ Follows label binding

*The node it's gonna
notify all the neighbors*

→ An LSR that has made a
binding, distributes it to
neighboring LSRs

→ Or at least to the upstream
one

Label Mapping

Association between

**Chosen by the
considered LSR**

→ **Input label**

→ **Output label**

**Chosen by
downstream LSR**

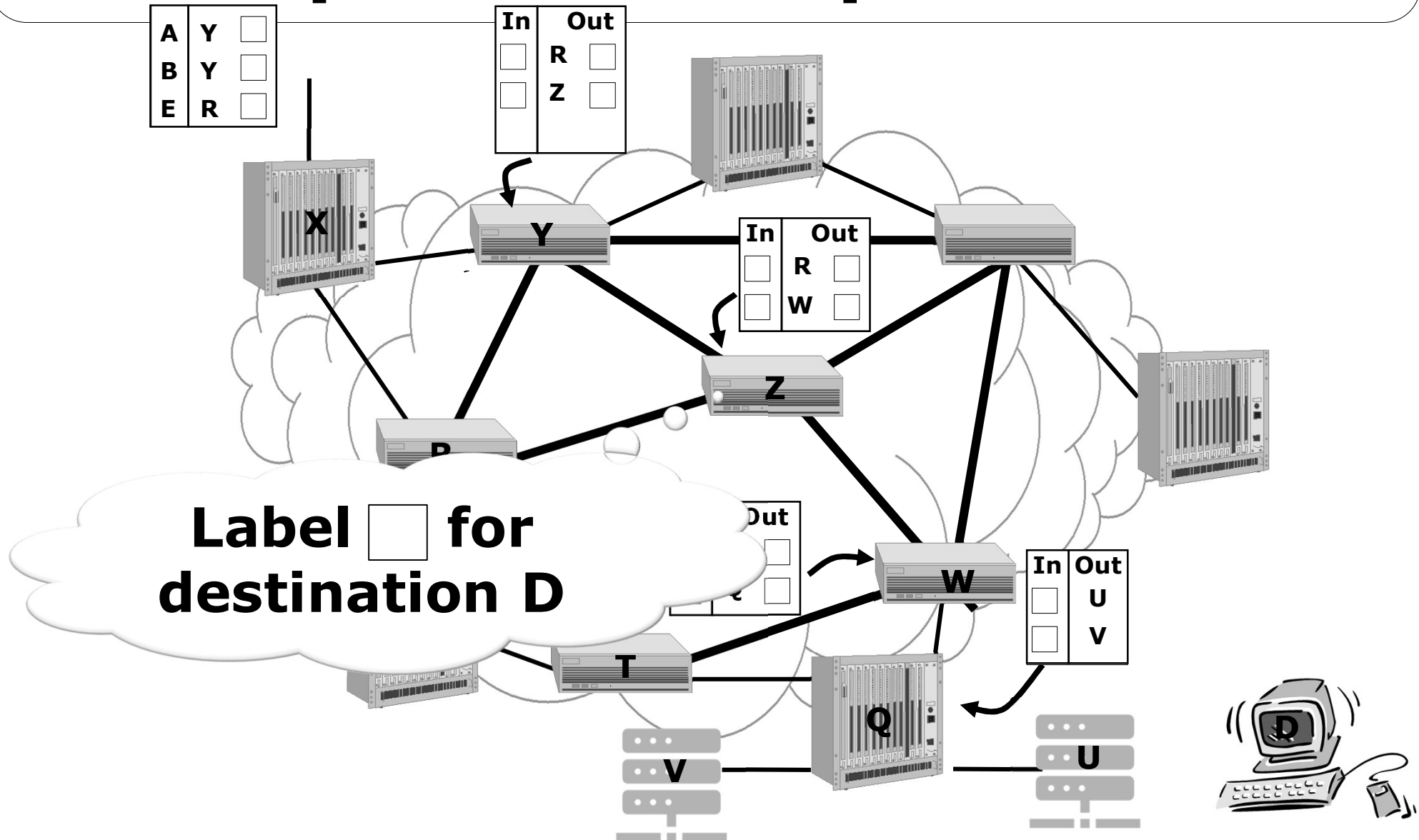
→ **Next hop**

Based on routing

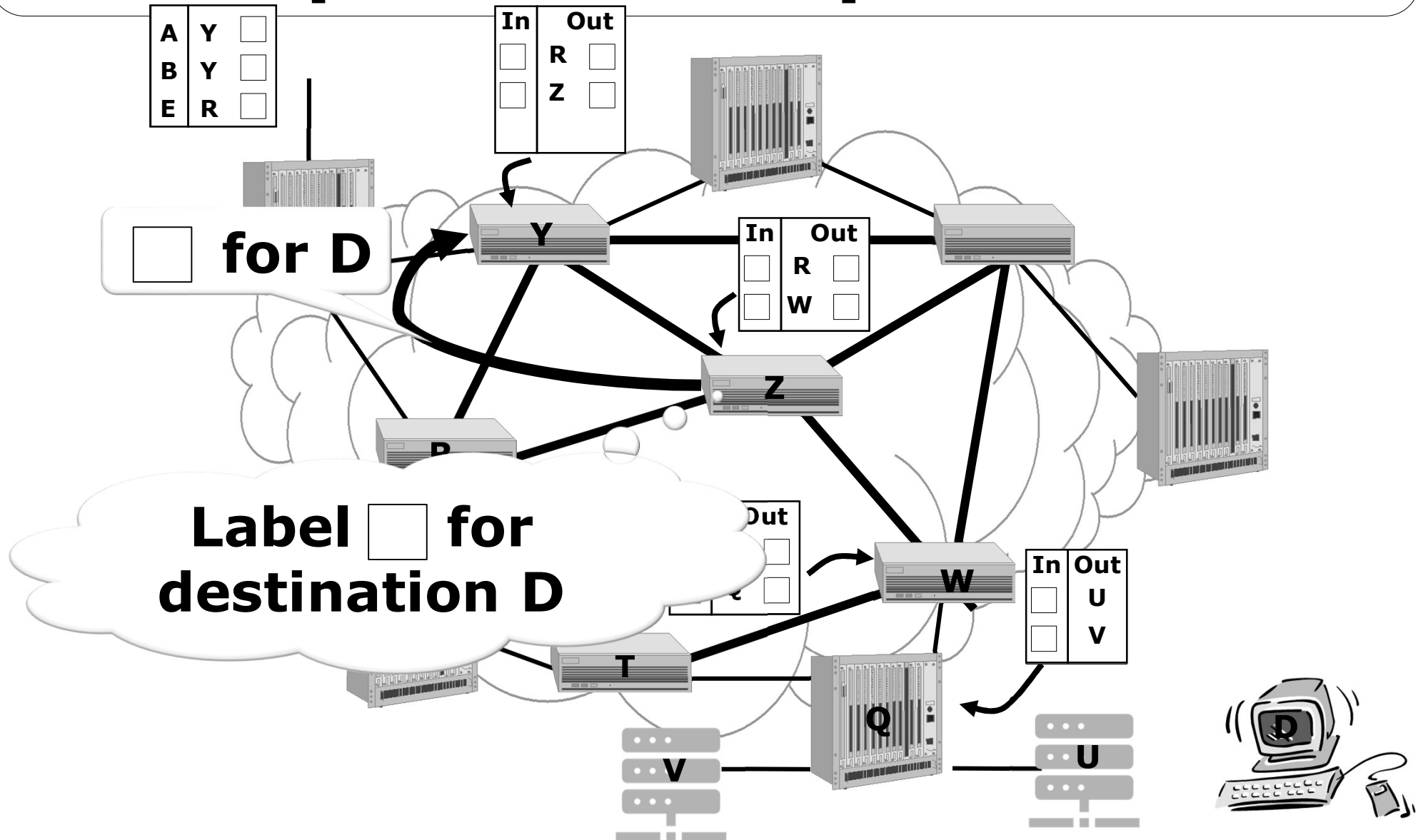
**How can the
LSR know it?**

Actual creation of an LSP

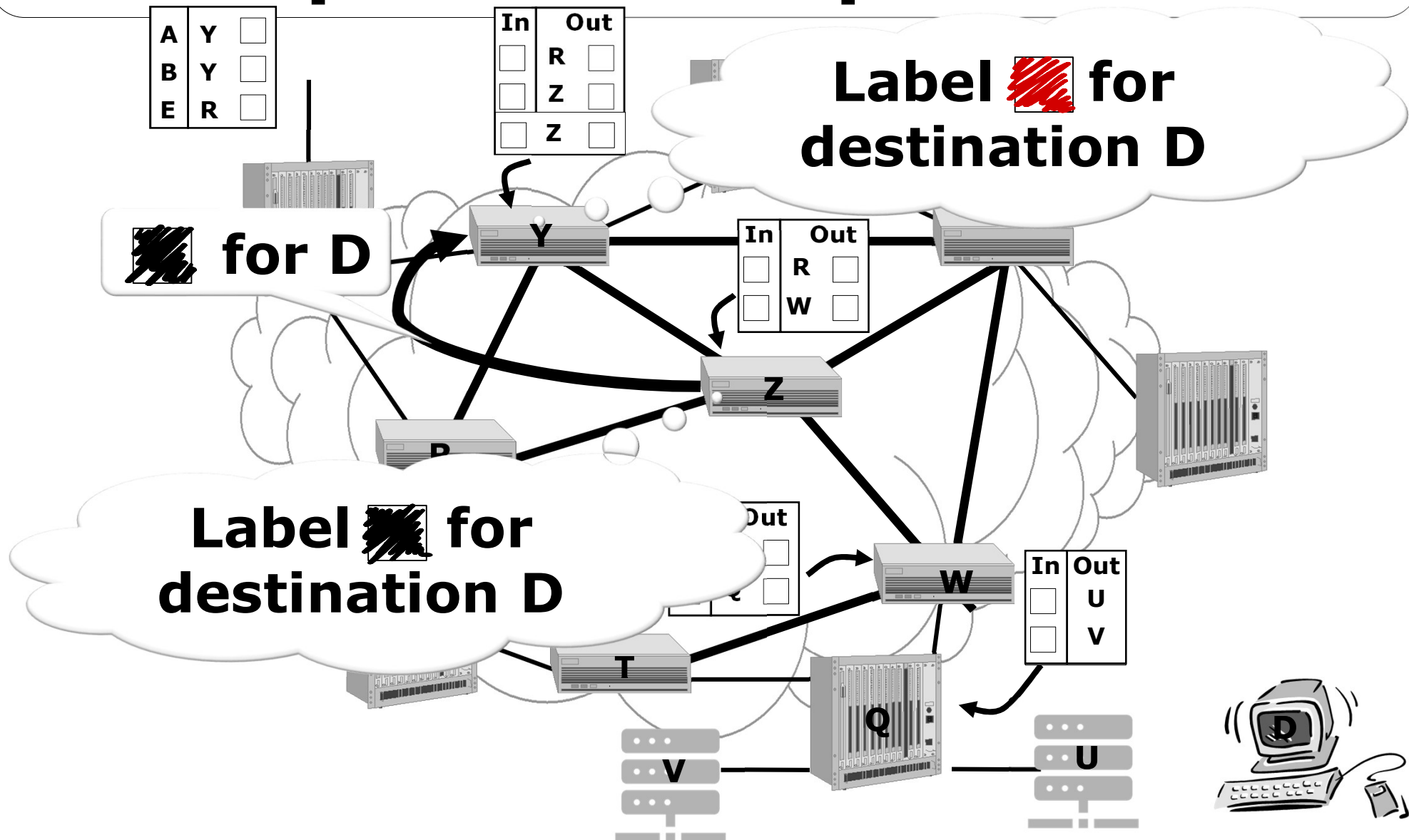
Putting it all together: Sample LSP Setup



Putting it all together: Sample LSP Setup



Putting it all together: Sample LSP Setup



Putting it all together: Sample LSP Setup

Distribution

A		
B	Y	
E	R	

In	Out
☒	R
☐	Z
☐	Z

Label ☒ for destination D

☐ for D

Label ☐ for destination D

Binding



Static label binding (and mapping)

- **Through management**
- **Equivalent to PVC ATM**
- **Non-scalable**
- **No interoperability
among managing systems**
- **Impossible to have LSPs
through different networks**

Dynamic Label Binding

→ **Data/traffic driven**

→ **Triggered by data packets**

→ **Control driven**

→ **Triggered by control message**

→ **Signaling**

→ **Routing**

Control Driven Label Binding

→ Topology based

- The creation of LSPs is linked to the discovery of routes towards destinations
- Unsolicited *whenever a new destination is discovered, a label is bounded*

→ Explicit creation of LSPs

- Explicit signaling
- Initiated by label edge routers
- On-demand

Label distribution protocols

Three alternatives (incompatible)

→ Routing protocol: BGP → Distance vector algorithm

→ Only topology based

→ Label distribution protocol (LDP)

→ Designed for the purpose

→ Resource reservation protocol (RSVP)

→ Designed for allocation in integrated service networks

"Please, be ready to receive this packet"

Routing protocols

Used to determine LSP routing

Impact label mapping phase

Indirectly determine packet routing

Routing protocols

Existing protocols

OSPF

IS-IS

Used today in most of the networks

BGP-4

carry topological information

**In MPLS context
they are enhanced to...**

Routing protocols

...carry information to constraint routing decisions (constraint data)

▶ **Capacity of links**

▶ **Link utilization** *We want to travel links that are not used so much*

▶ **Dependencies among links**

▶ **Used for fault recovery**

Enhanced routing protocols

**Constraint based routing
is fundamental to support
traffic engineering**

OSPF-TE

IS-IS-TE

Routing Modes

Hop-by-hop routing

Explicit routing

Hop-by-hop Routing

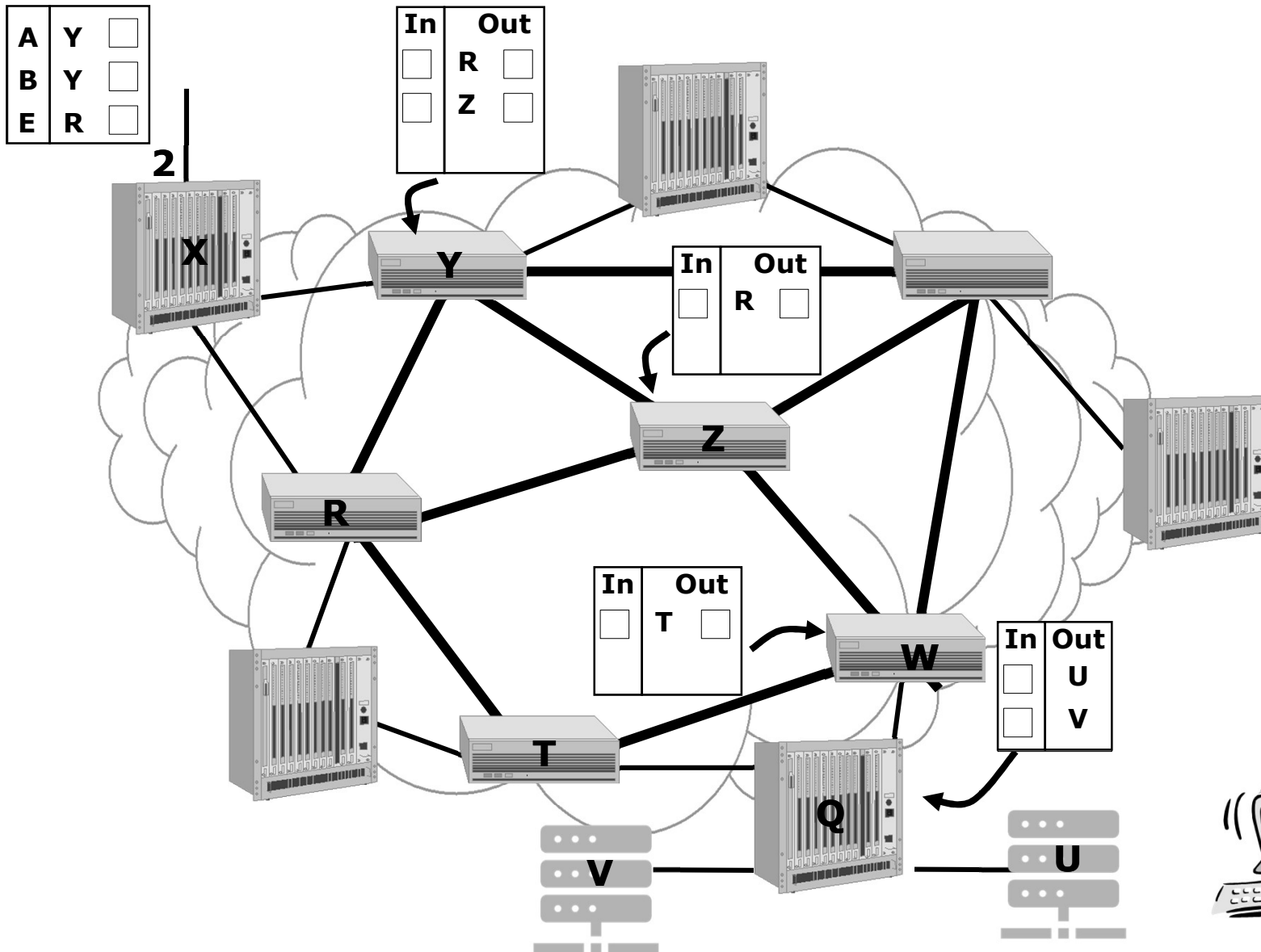
Each LSR decides the next LSR of the LSP path

- **E.g., Based on IP routing table**
- **Same principle and outcome (route) as in traditional IP routing**

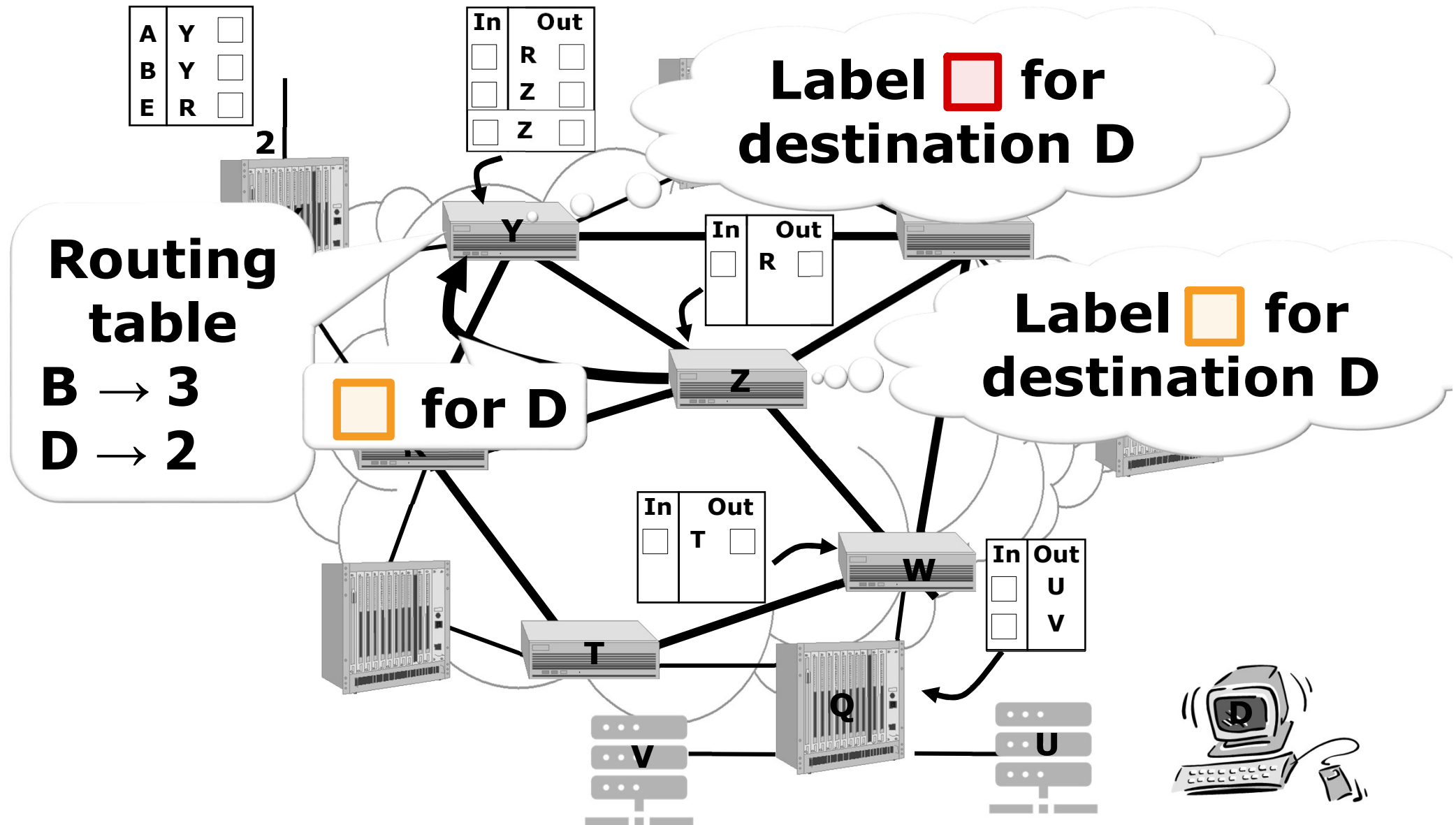
Hop-by-hop Routing Procedure

- **Pick label for upstream link**
 - **Label binding**
- **Map it to**
 - **Next hop**
 - **Address of connected interface of next LSR**
 - **Label announced by next LSR**

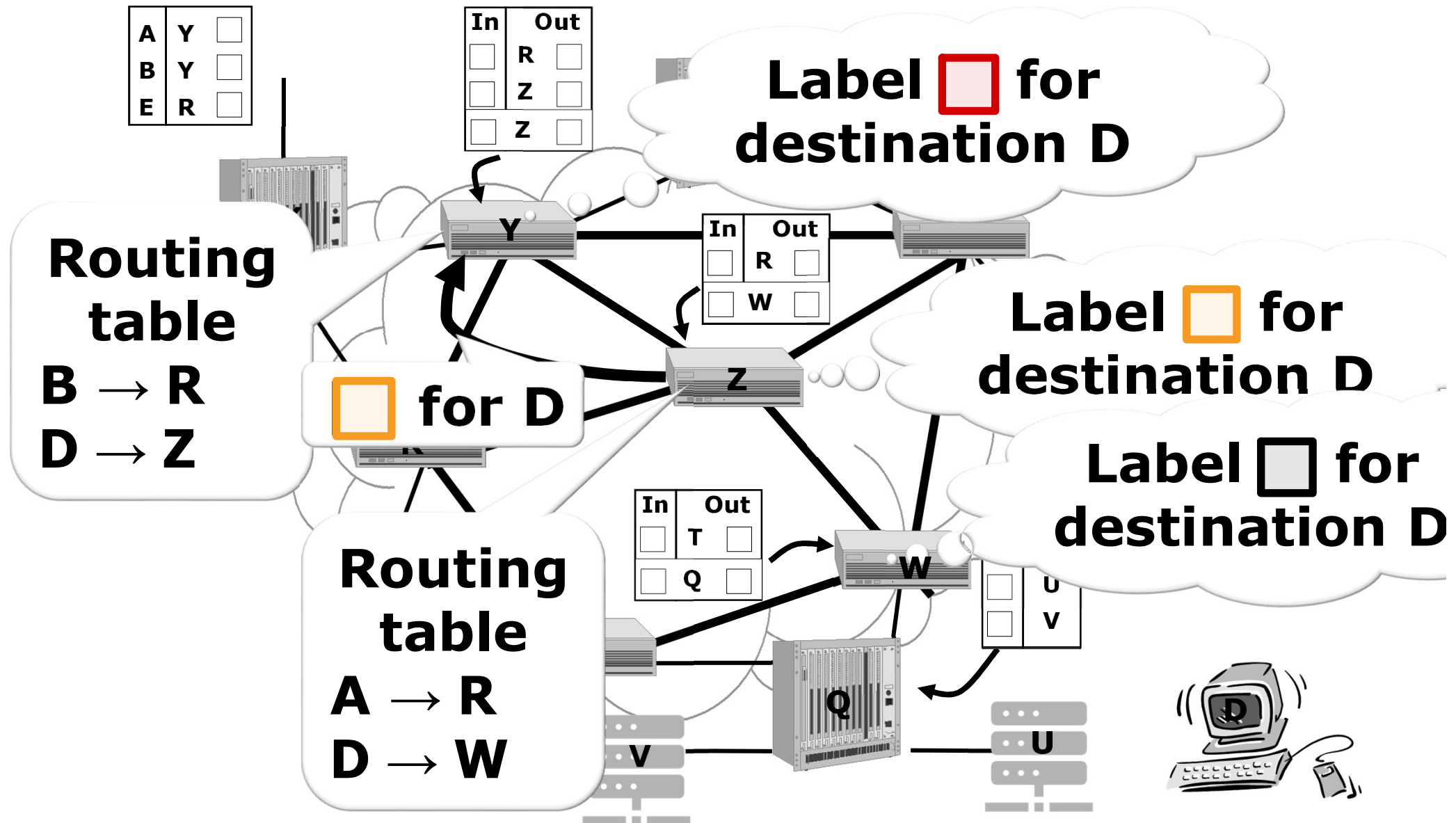
Hop-by-hop Routing Example



Hop-by-hop Routing Example



Hop-by-hop Routing Example



Explicit constraint based routing

→ **A single switch chooses the path of a whole LSP**

→ **For example, ingress LSR**

→ **Explicit routing**

→ **The choice is constraint based**

→ **Constraint based routing**

Constraint Based Routing

Distributed operation impossible

→ No unique route selection criteria

→ Conflicting constraints

→ Hard to maintain constraint information synchronized

→ It changes more frequently than topology information

Label distribution protocols

Should be modified

→ **CR-LDP**

→ **Constraint-based routing LDP**

→ **RSVP-TE**

→ **RSVP for Traffic Engineering**

**To be used with
OSPF-TE and IS-IS-TE**

New possibilities

→ **Traffic engineering**

→ **Guaranteed quality of service**

IP was not developed for QoS

→ **Not yet supported**

MPLS do not solve the problem but makes solving the problem easier

→ **Per-class traffic engineering**

→ **Synergy with DiffServ**

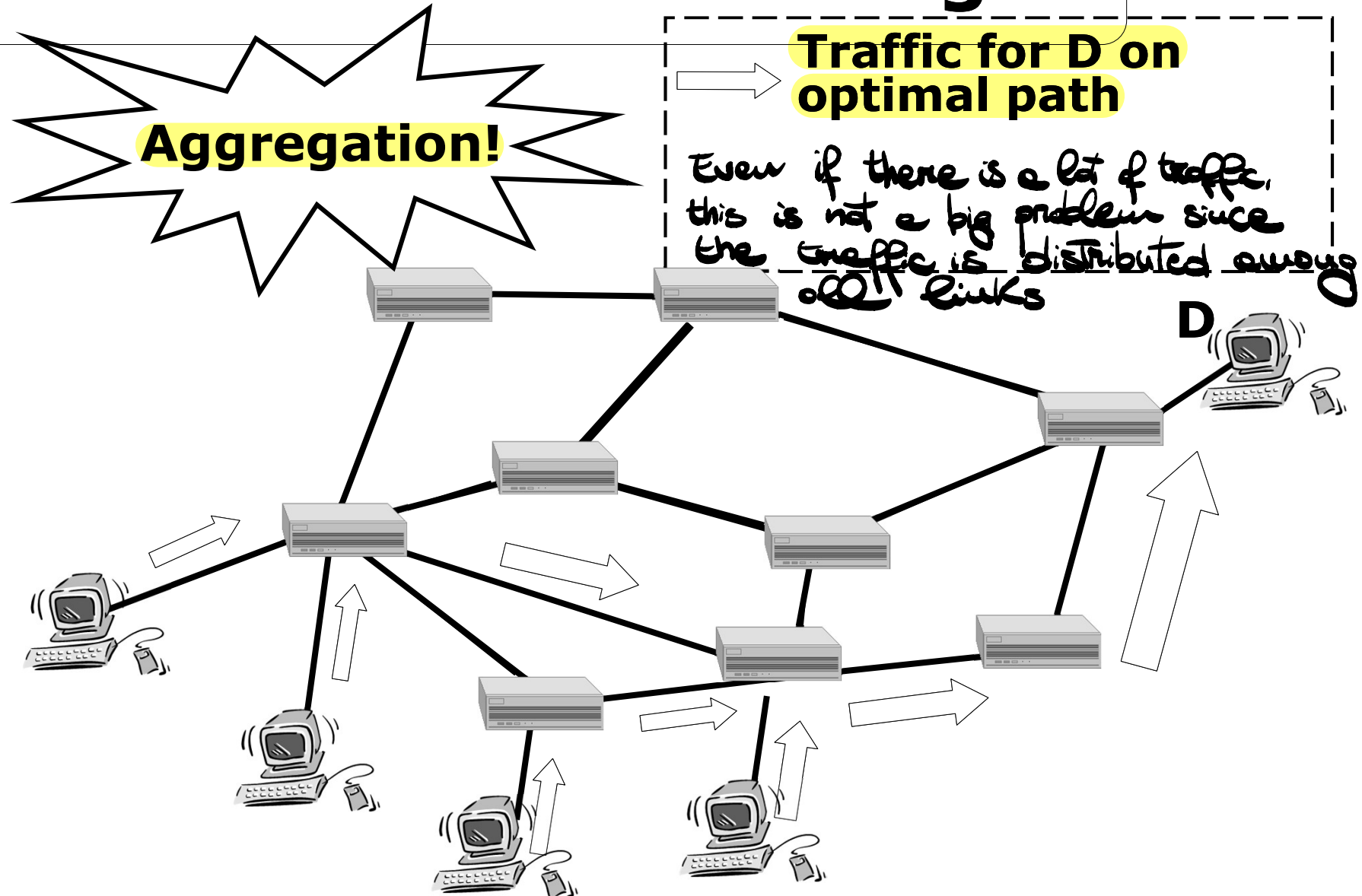
→ **Fast fault recovery**

→ **In less than 50 ms**

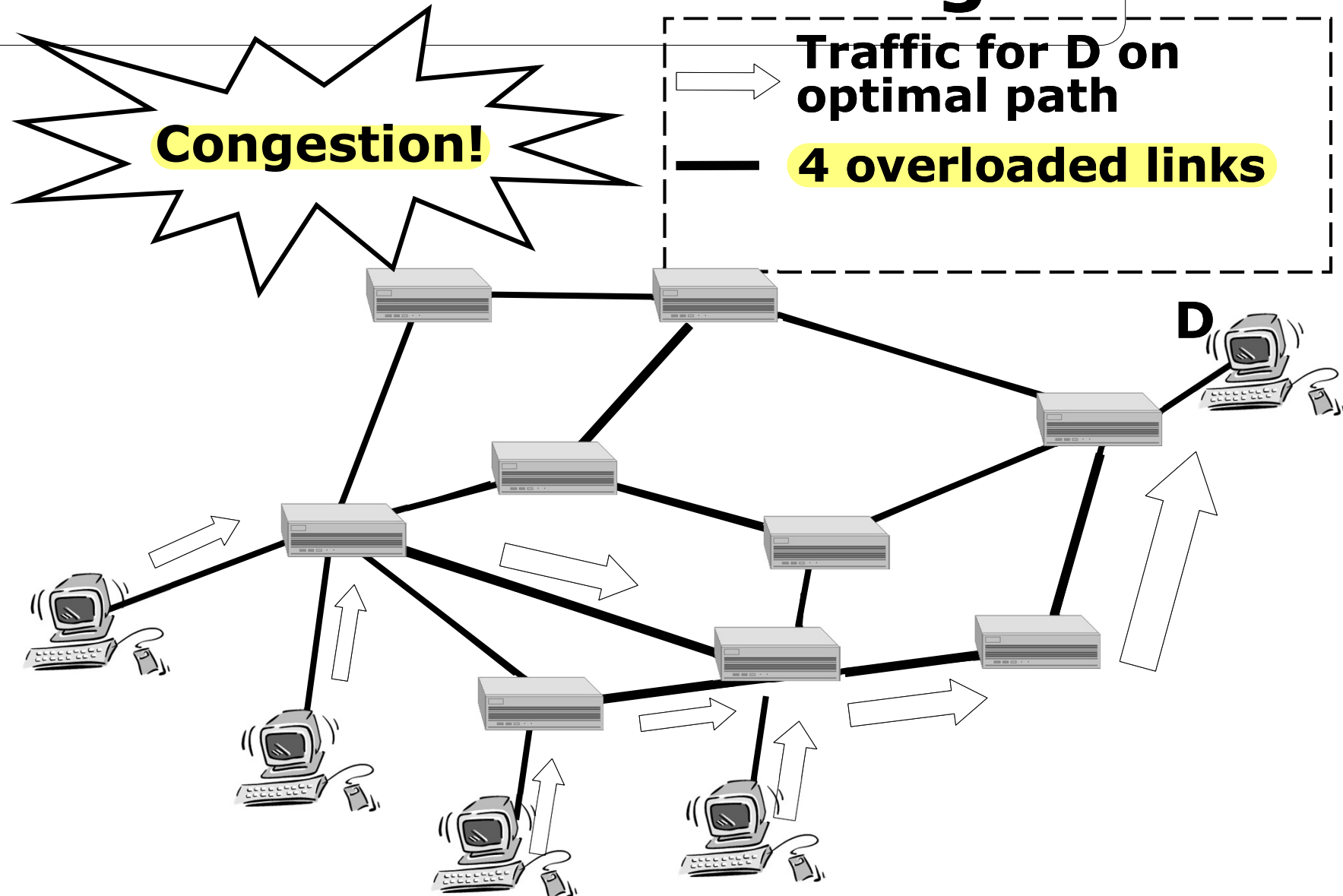
Traffic Engineering

Motivation and enablement

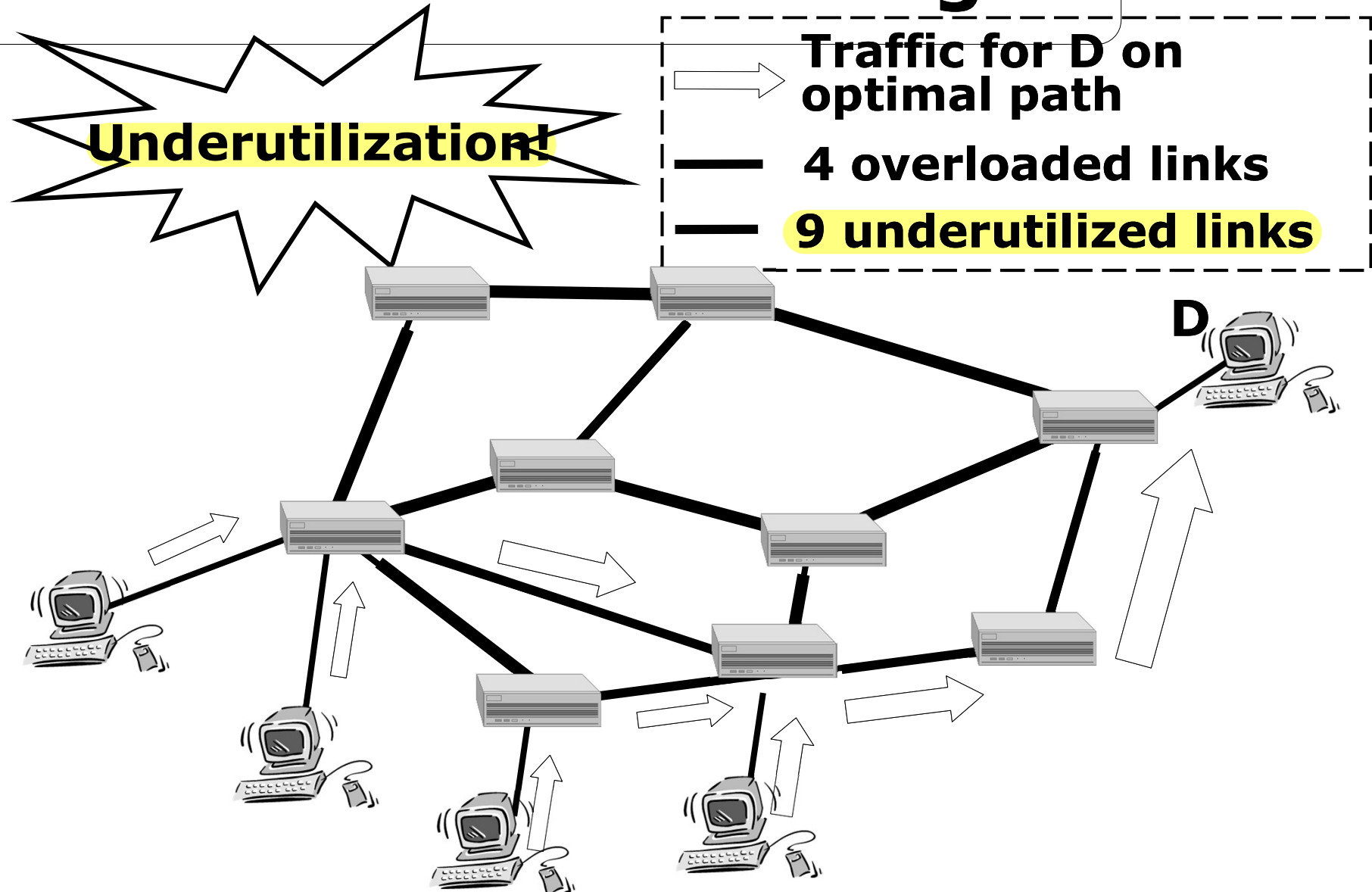
Traditional IP routing



Traditional IP routing

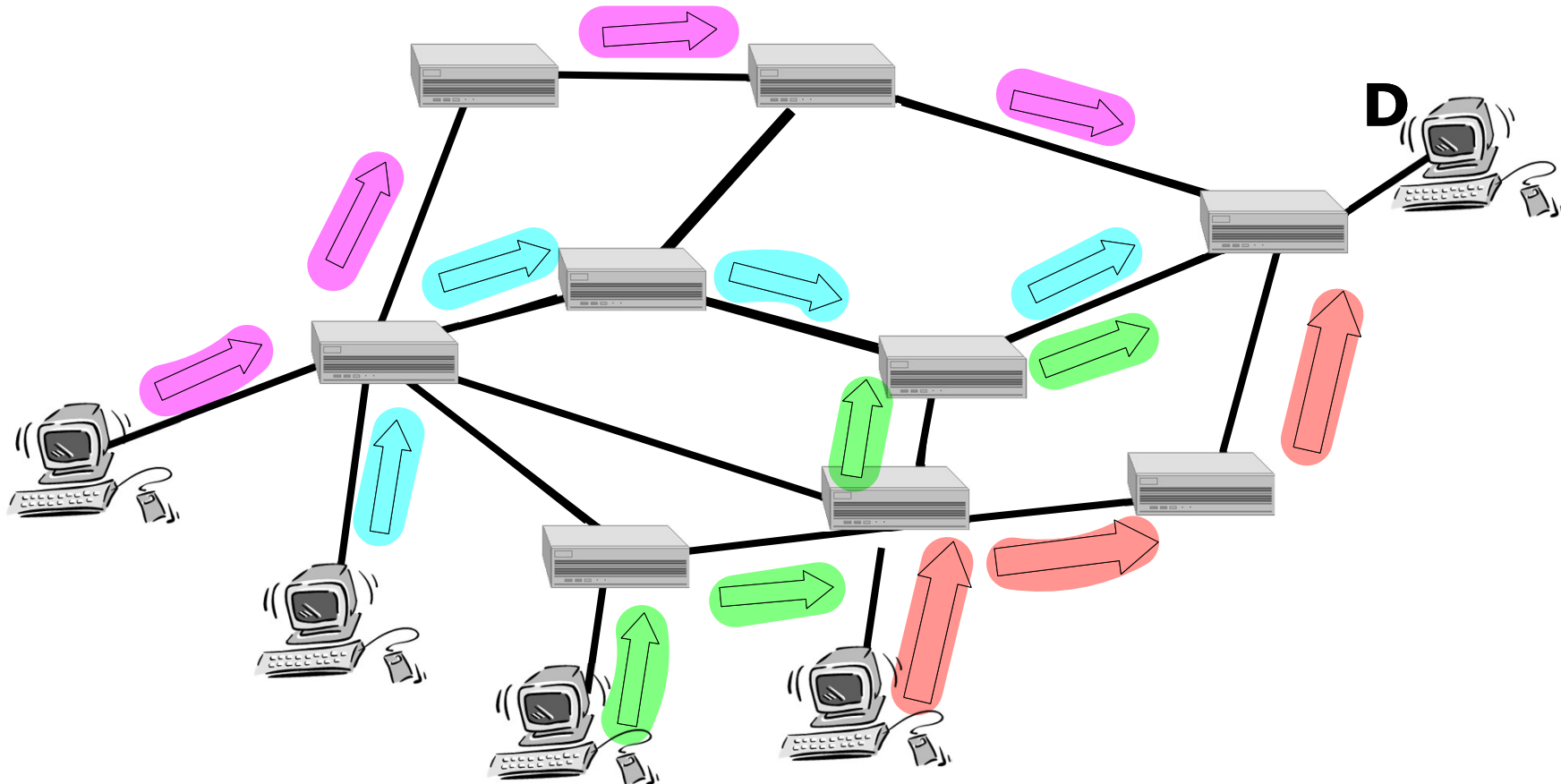


Traditional IP routing

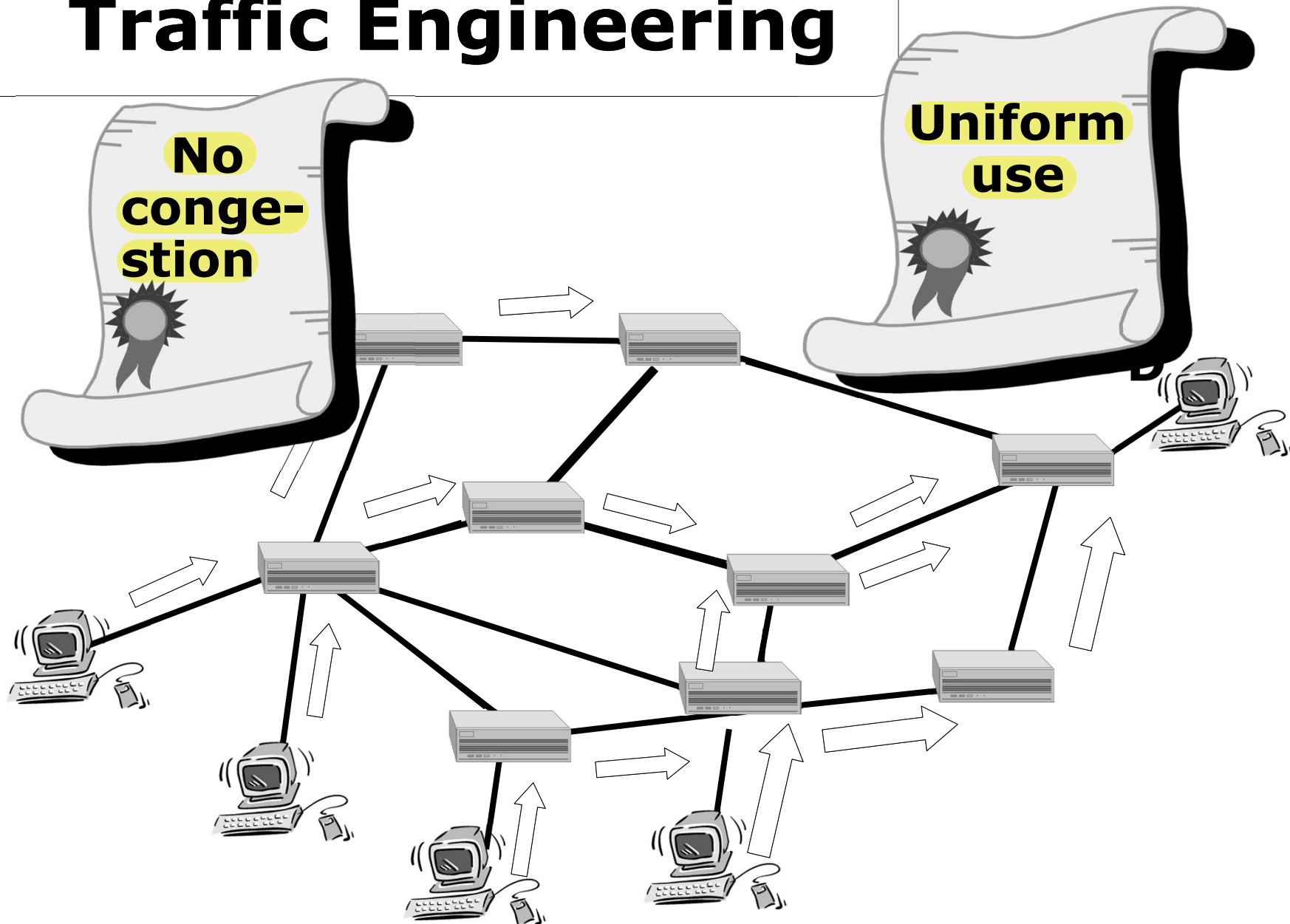


Traffic Engineering

Enables traffic distribution



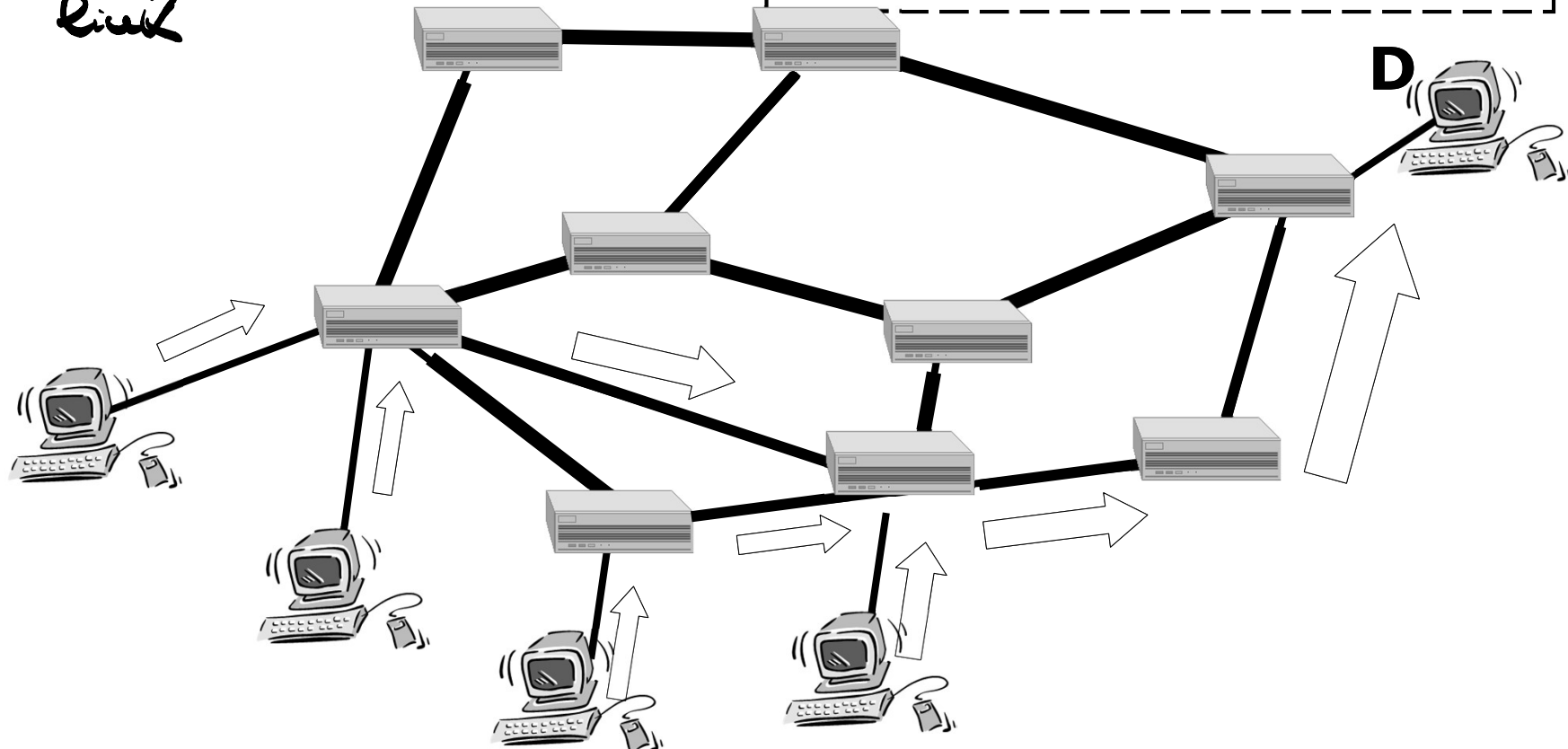
Traffic Engineering



Can you do it with traditional IP routing?

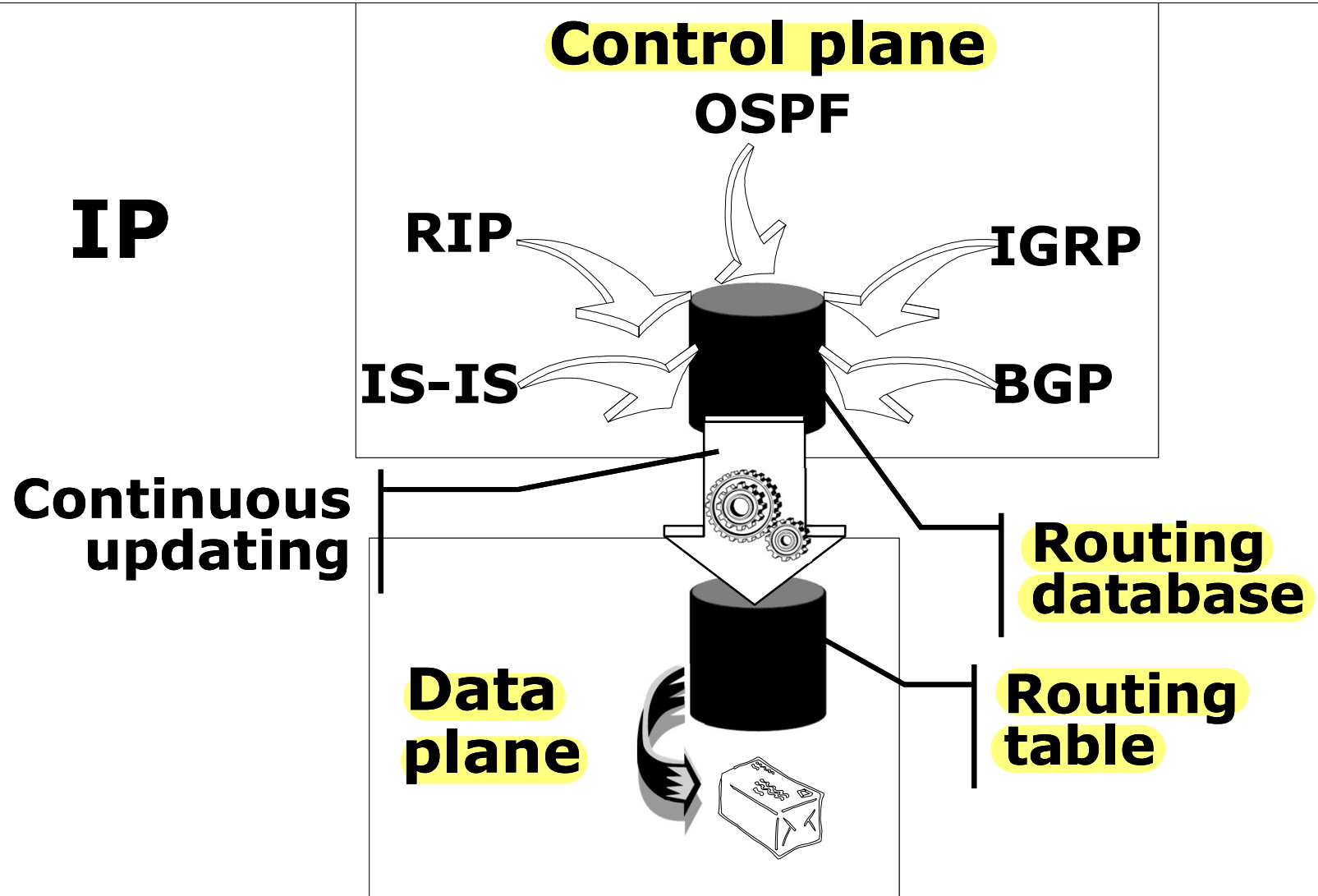
No because the router would send the whole traffic on that underloaded link

→ Traffic for D on optimal path
— 4 overloaded links
— 9 underutilized links

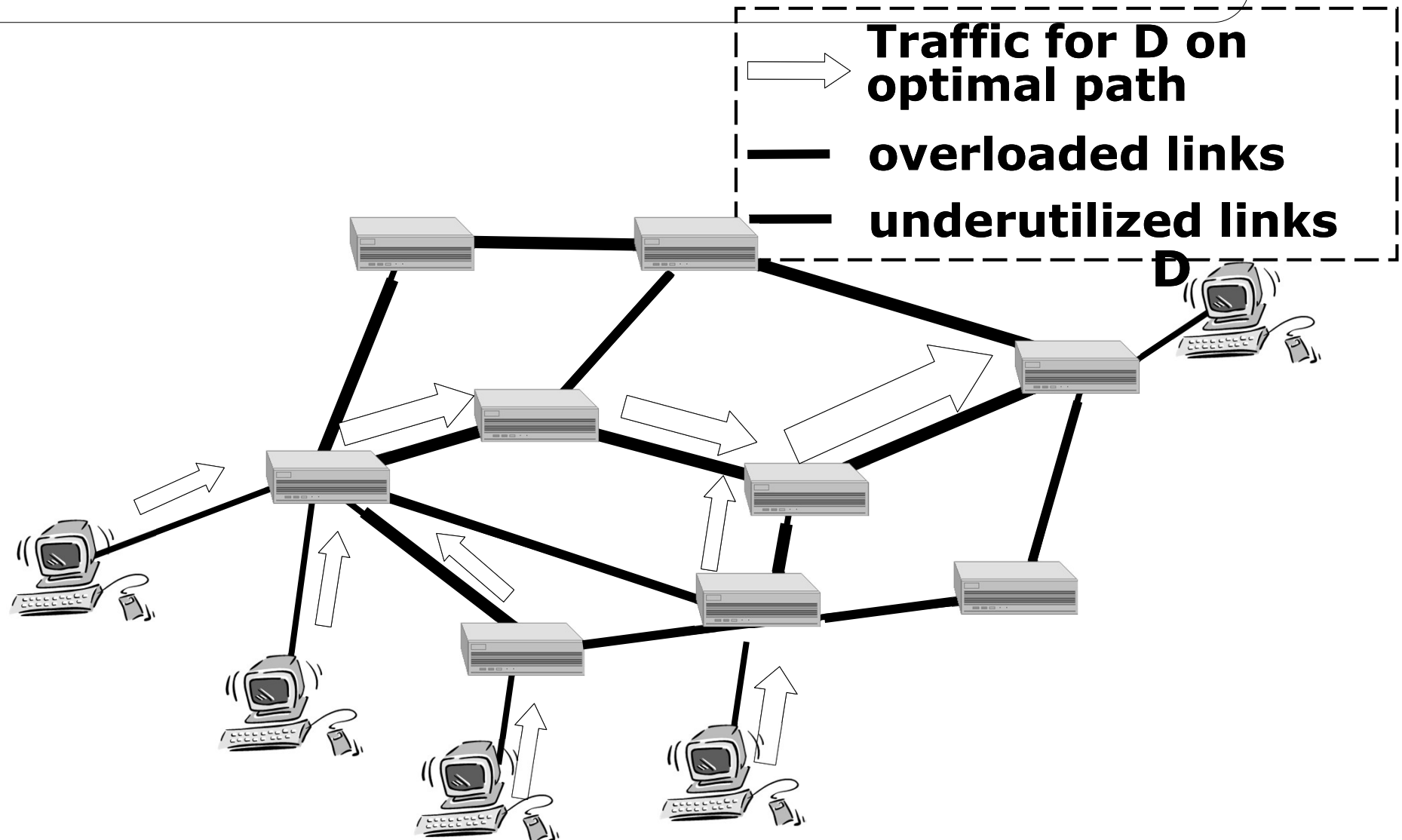


Control plane and data plane

R
O
U
T
E
R
I
P

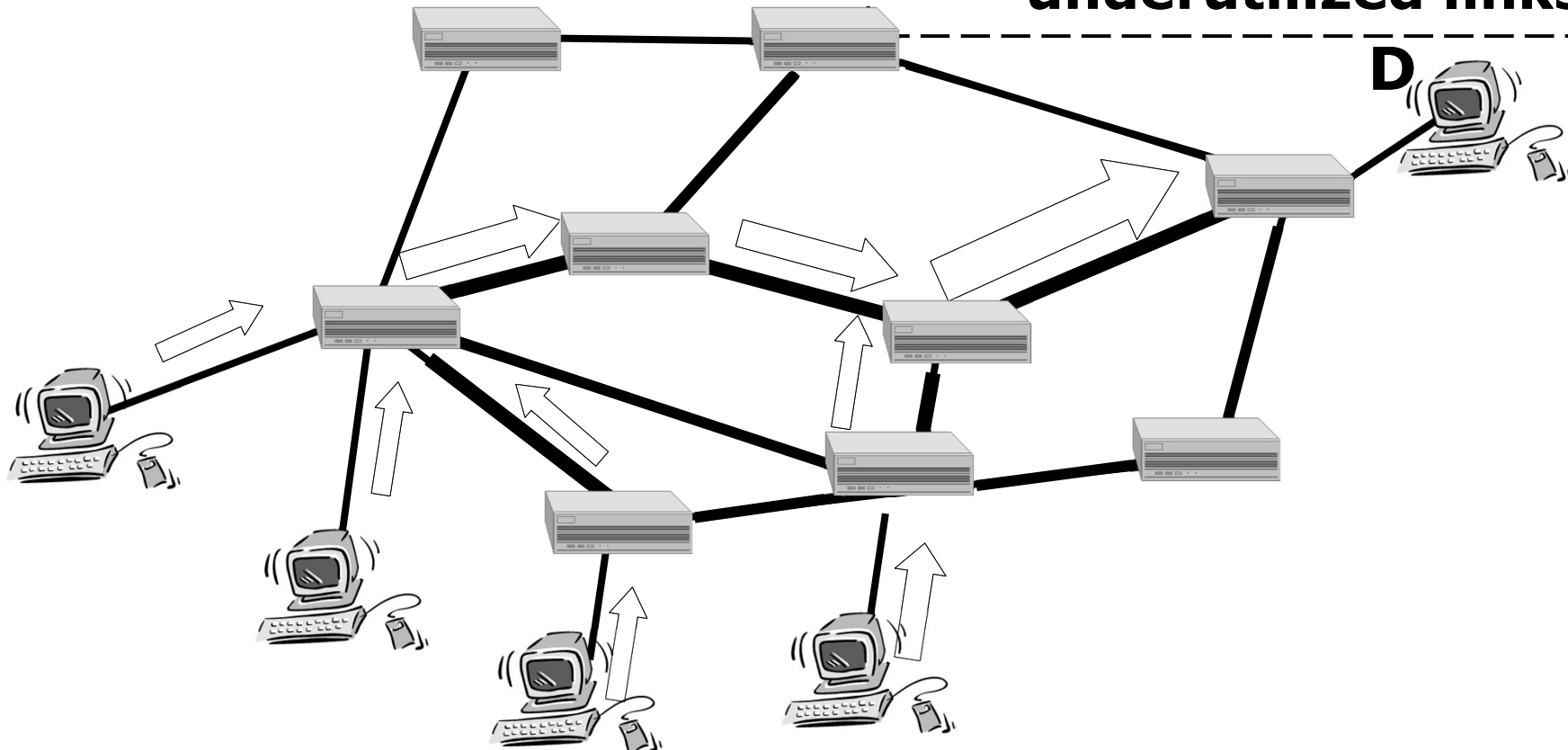


It would be possible to choose paths according to load

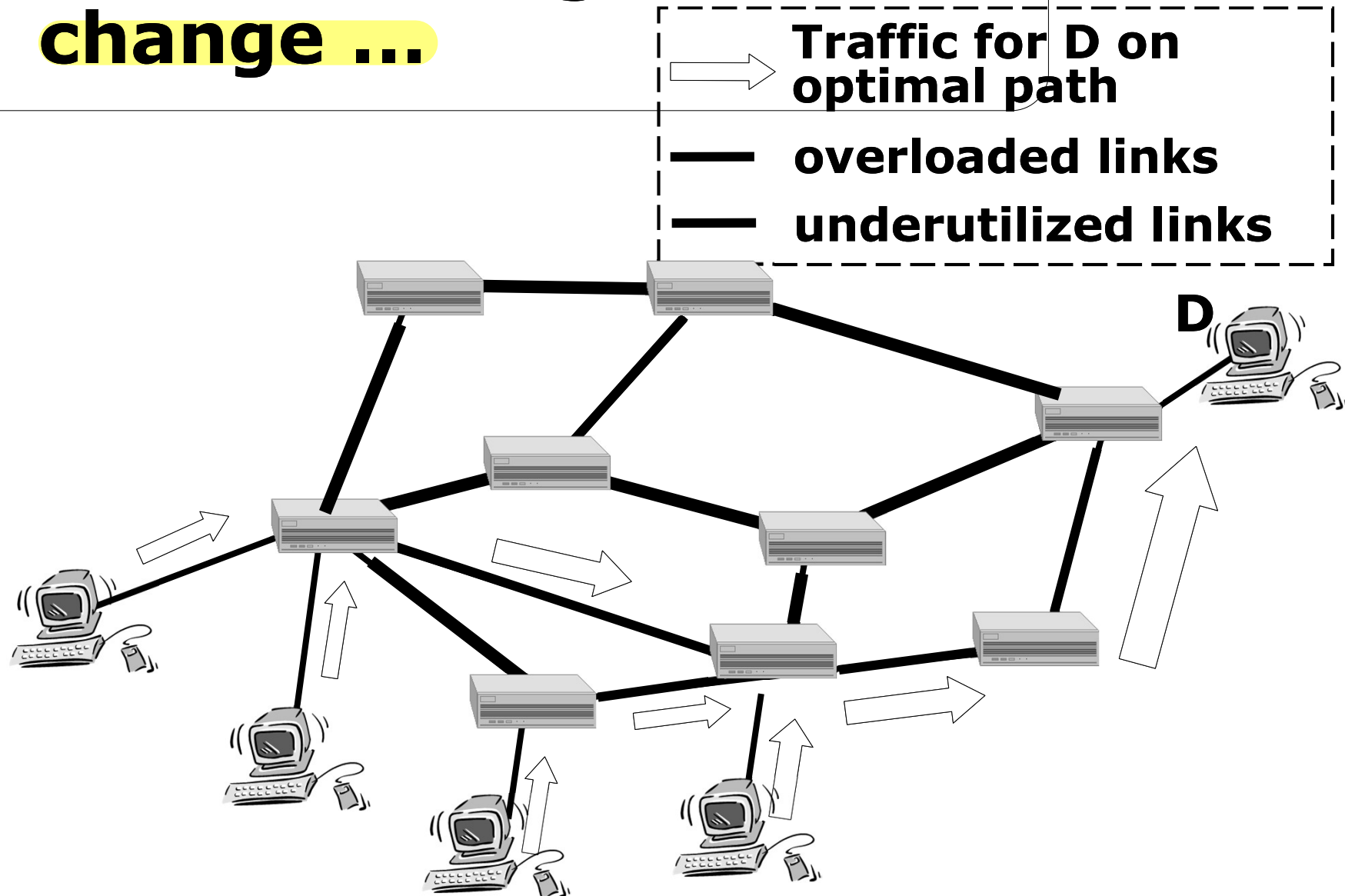


In such way, unloaded links are loaded and viceversa ...

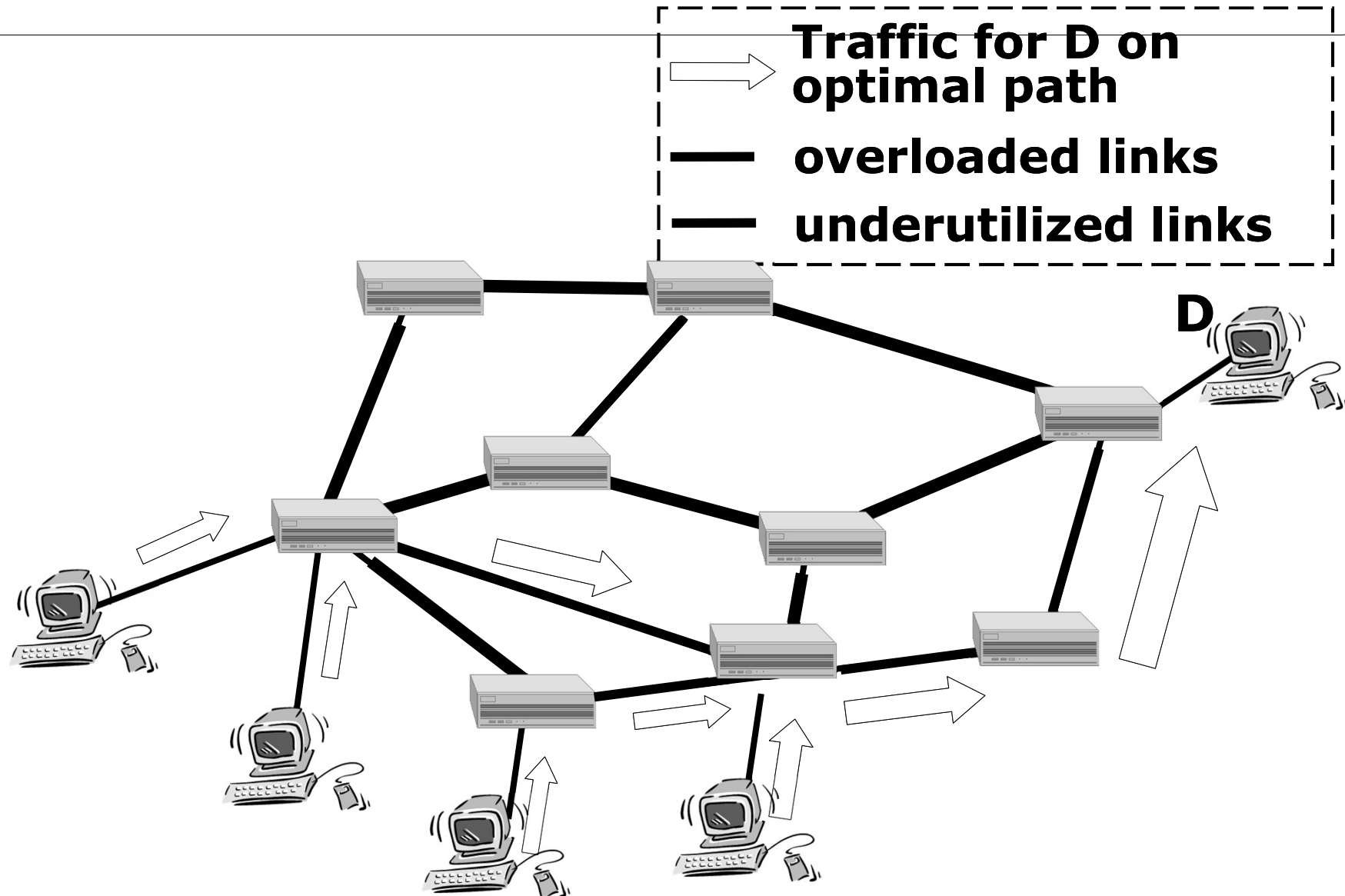
→ Traffic for D on optimal path
— overloaded links
— underutilized links



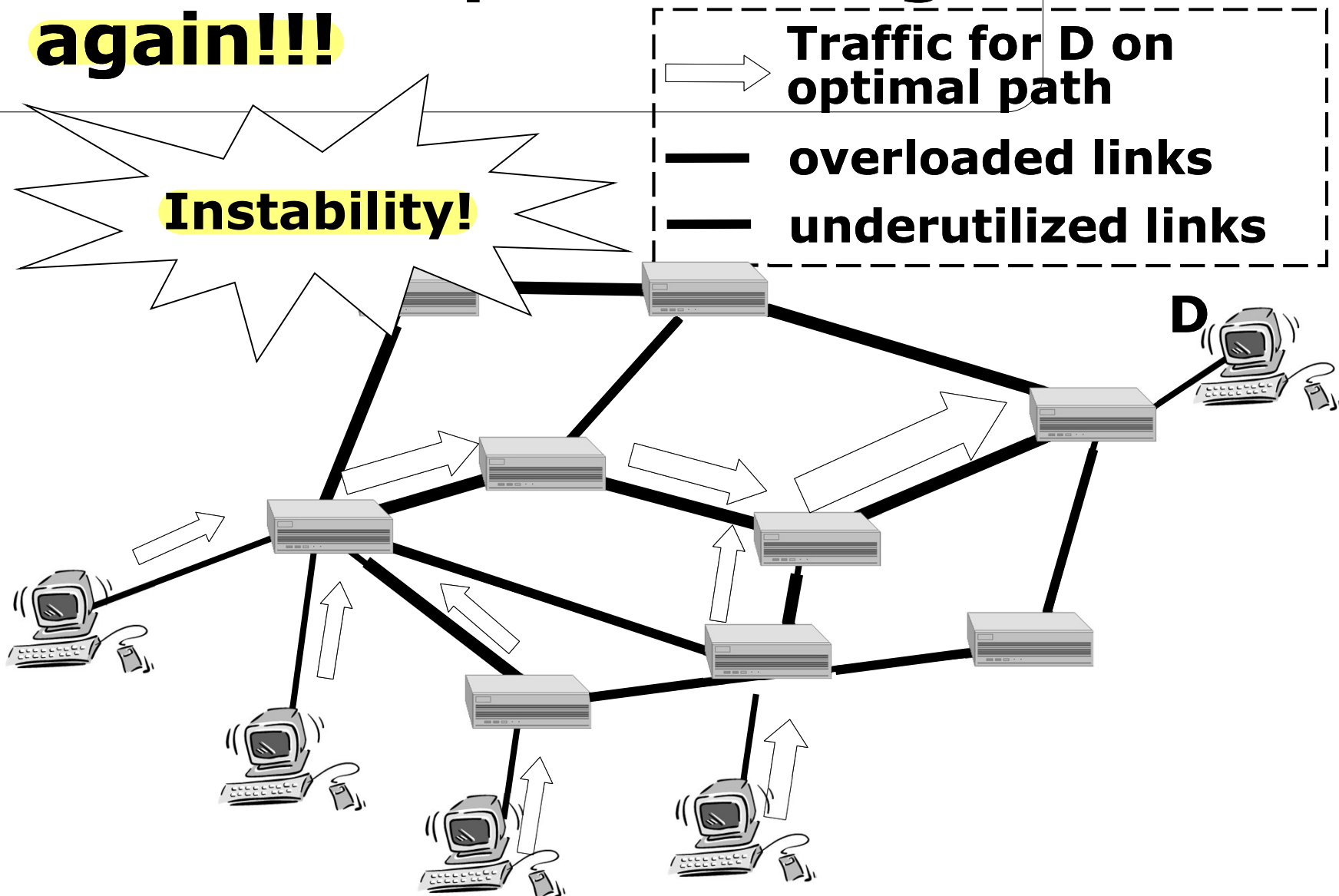
... and routing tables change ...



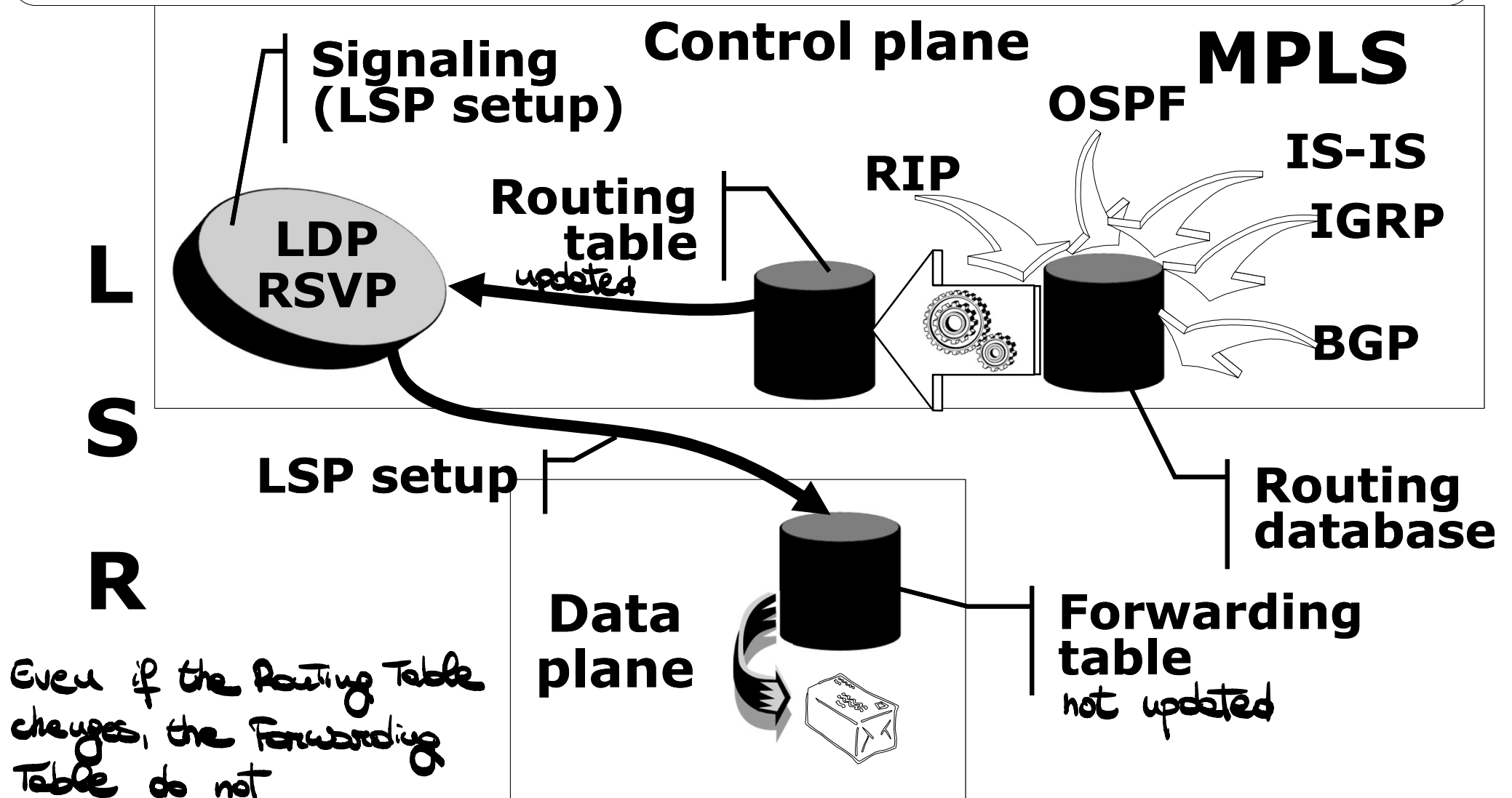
...so does the link load...



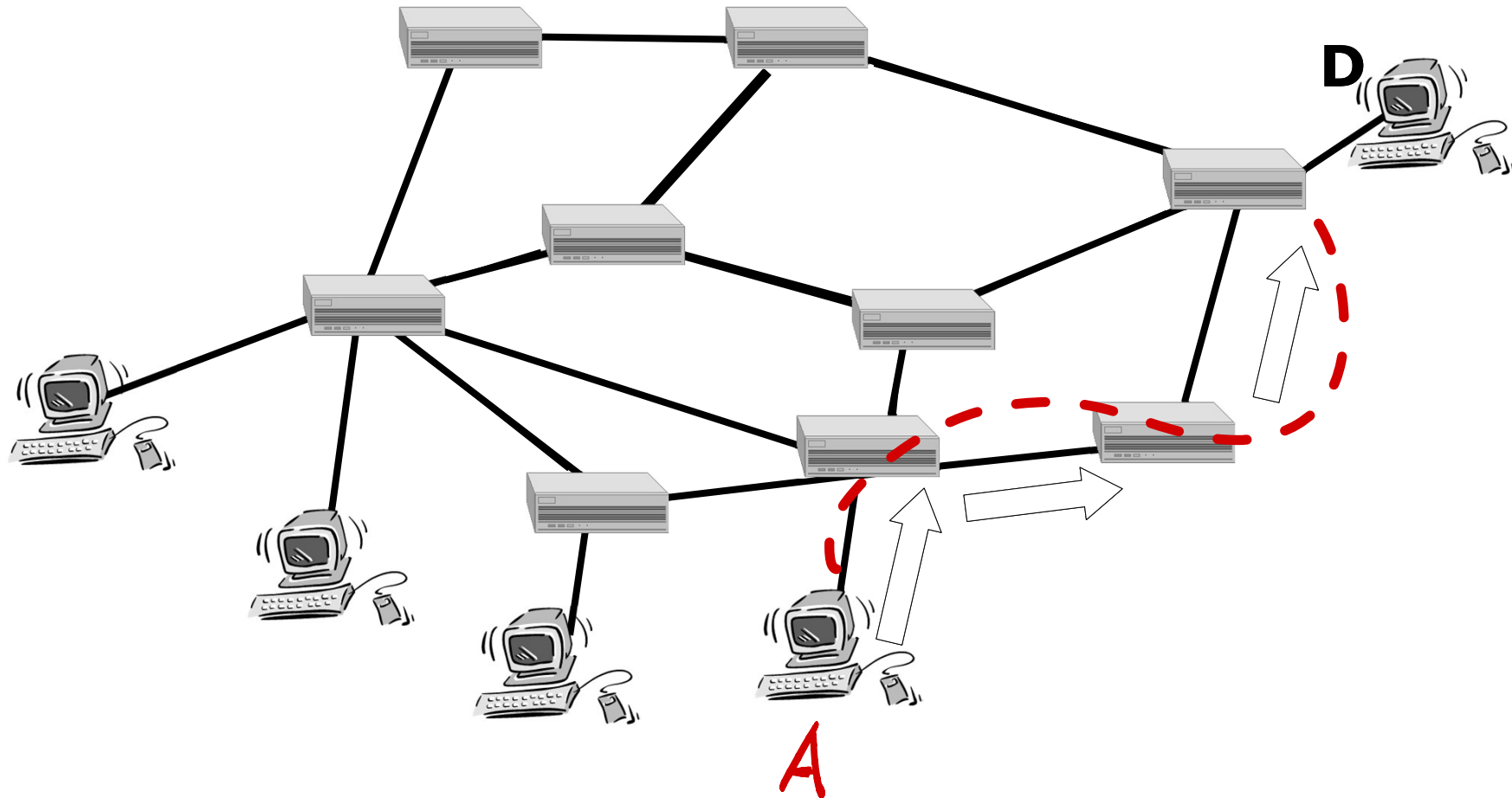
... and the paths change again!!!



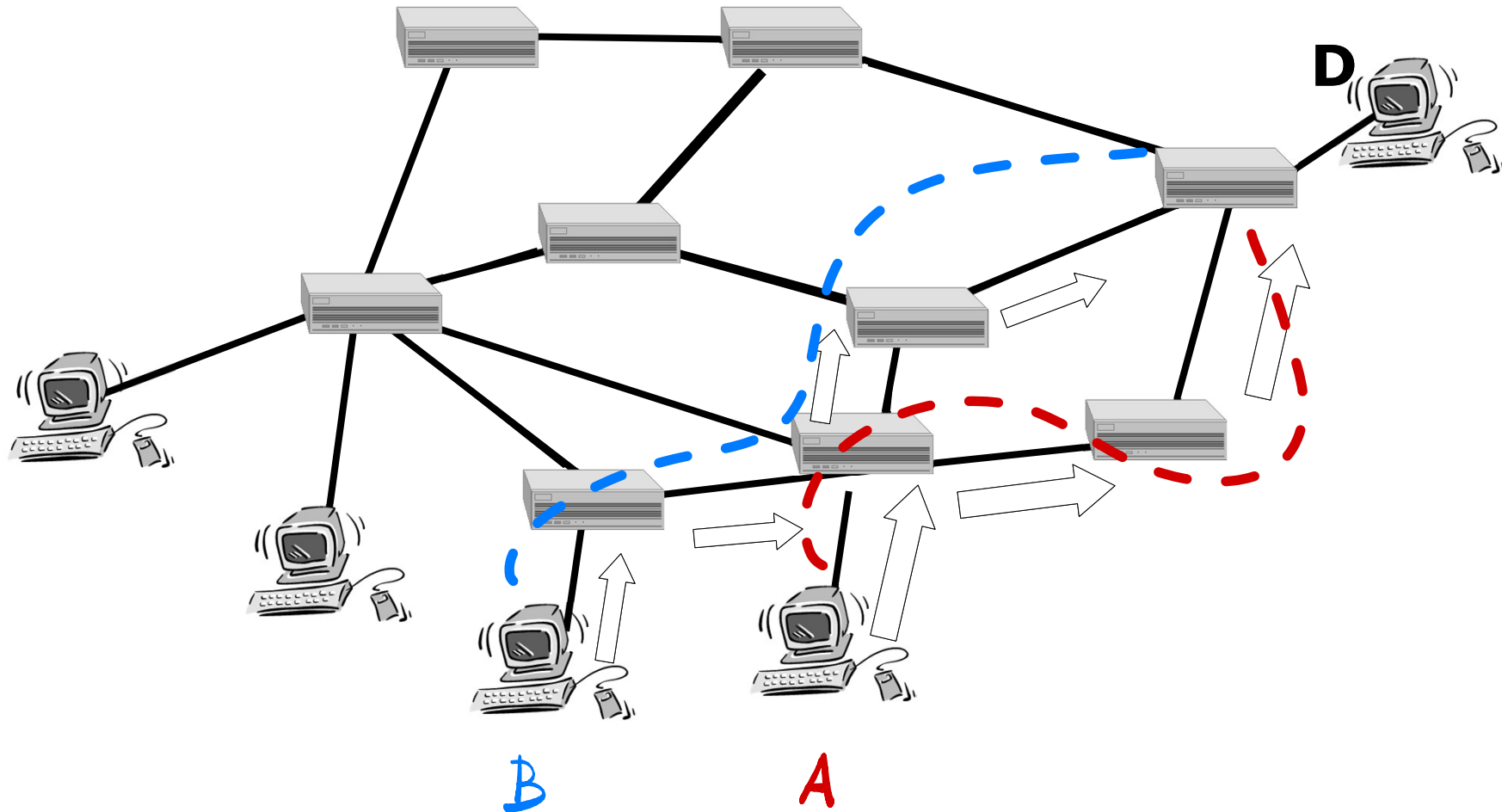
Control plane and data plane



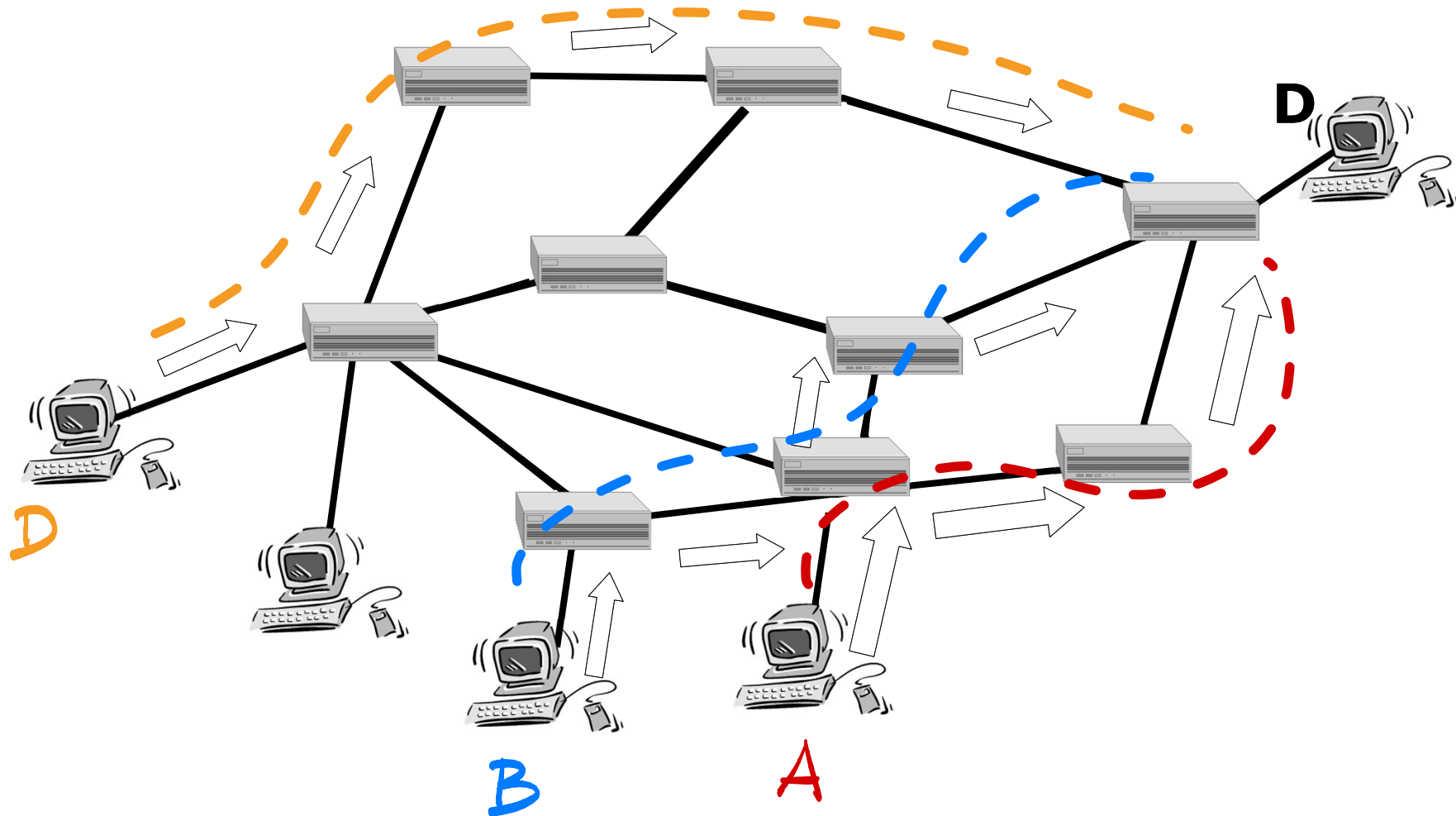
Traffic Engineering with MPLS



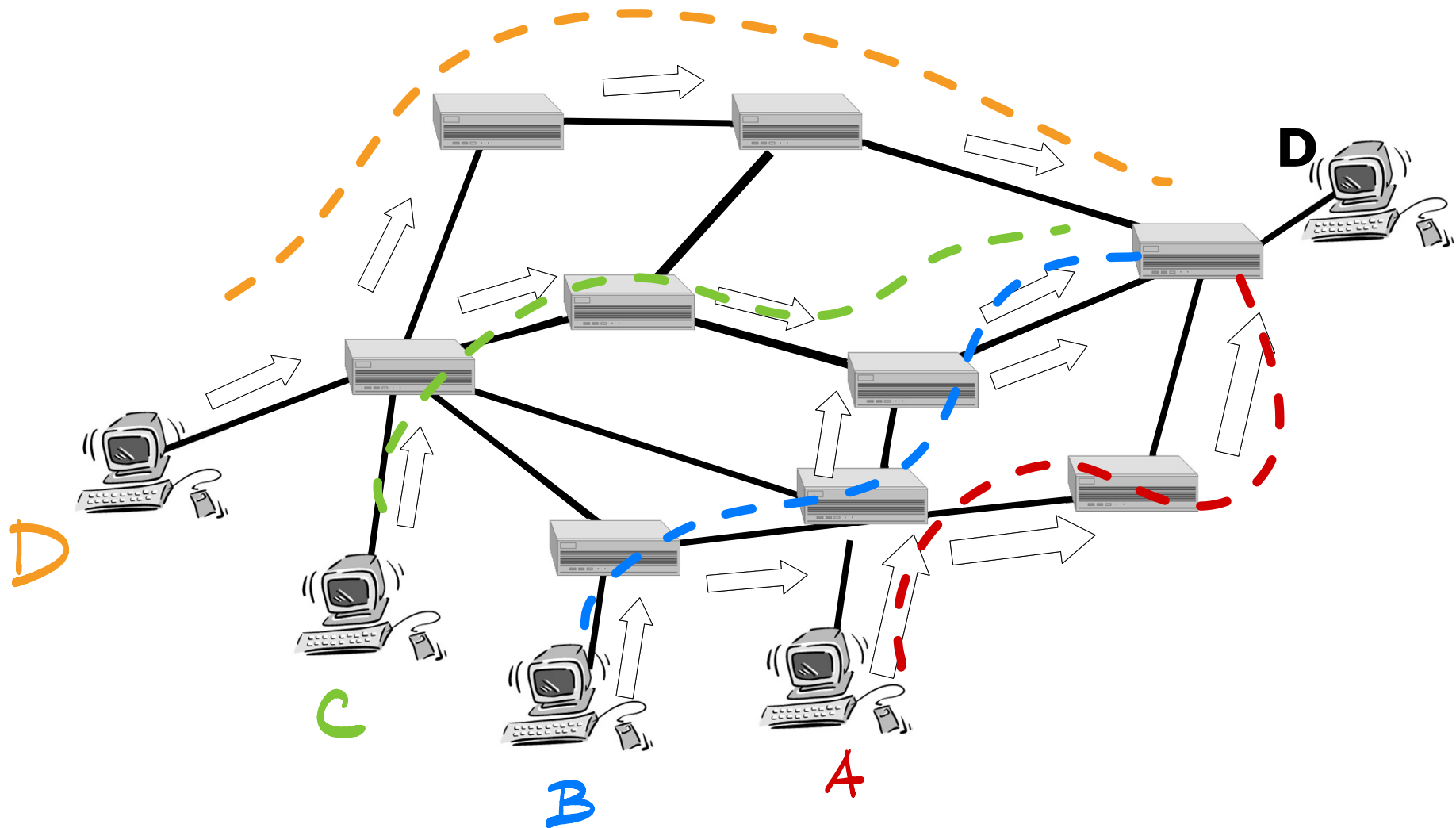
Traffic Engineering with MPLS



Traffic Engineering with MPLS



Traffic Engineering with MPLS



Traffic Engineering without MPLS

- **ATM has so far been used for traffic engineering on IP networks**
- **Two control plans**
 - **Routers are ATM-unaware**
- **Great number of adjacencies**
 - **Limited scalability**

Traffic Engineering with MPLS

MPLS is IP-aware

**Only one control plan operating
on physical topology**

→Simpler

→Greater scalability

MPLS extensions

- **MPλS (MPLambdaS)**
- **MPLS control plans in optical networks**
- **GMPLS (Generalized MPLS)**
- **MPLS control plane in any network**
 - **Packet, circuit, optics, etc...**

Adding CoS and QoS

**Resources and service modes
may be associated to a FEC at
LSP setup**

**Explicit support is required
in LSR data plan and control
plan**

Class of service (CoS)

- Relative priority among different FECs**
- It does offer absolute guaranties**
- Support of DiffServ model**
 - Per-hop behavior**
 - EF (expedite forwarding), AF (assured forwarding)**
- Per class traffic engineering**
 - DS-aware traffic engineering**

Quality of service (QoS)

Specific guaranties on

→ Bandwidth

→ Delay

→ Burst size

Advantages of QoS in MPLS

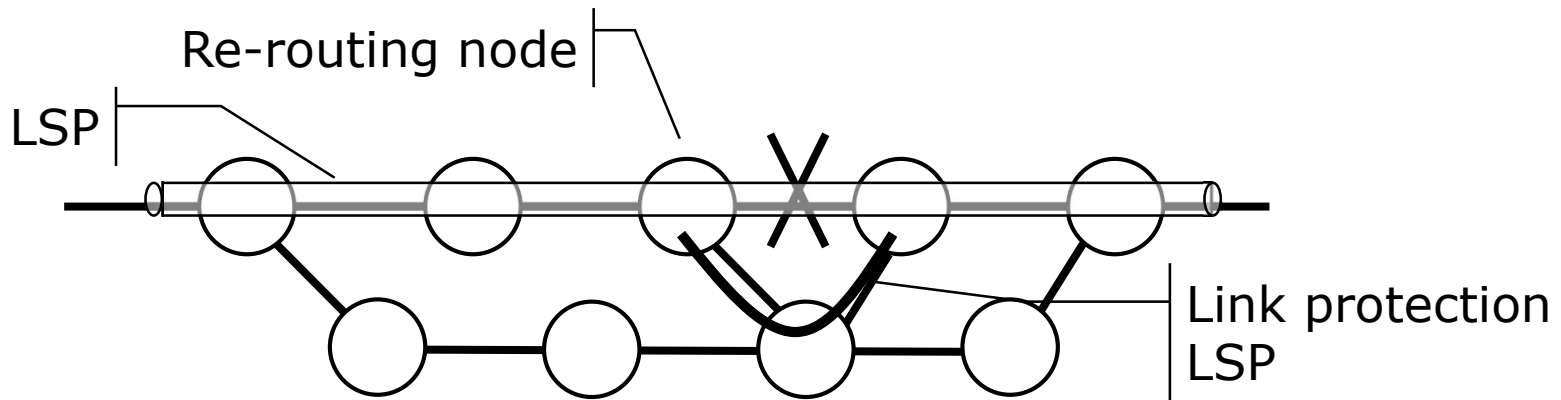
Unified network supporting all types of services

(marketing message)

- Support of QoS and real-time services on IP (e.g., voice) is not ready**
- A lot of multi-service networks now have a paradigm “Ships-in-the-night”**
 - ATM protocols are ATM typical services**
 - MPLS control plan for IP services**

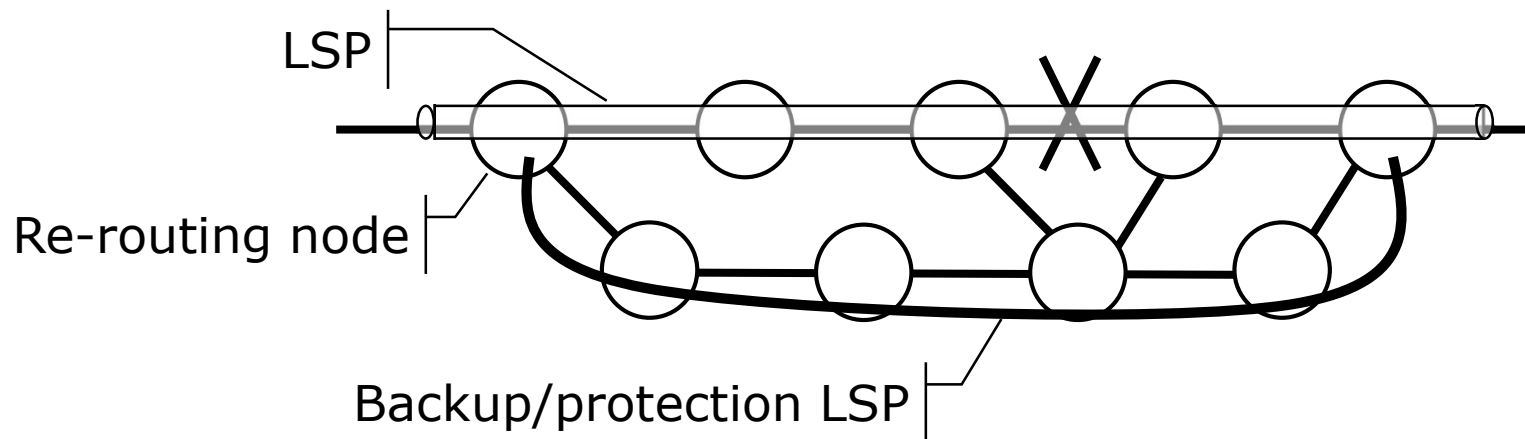
Fast Fault Recovery

Link re-routing

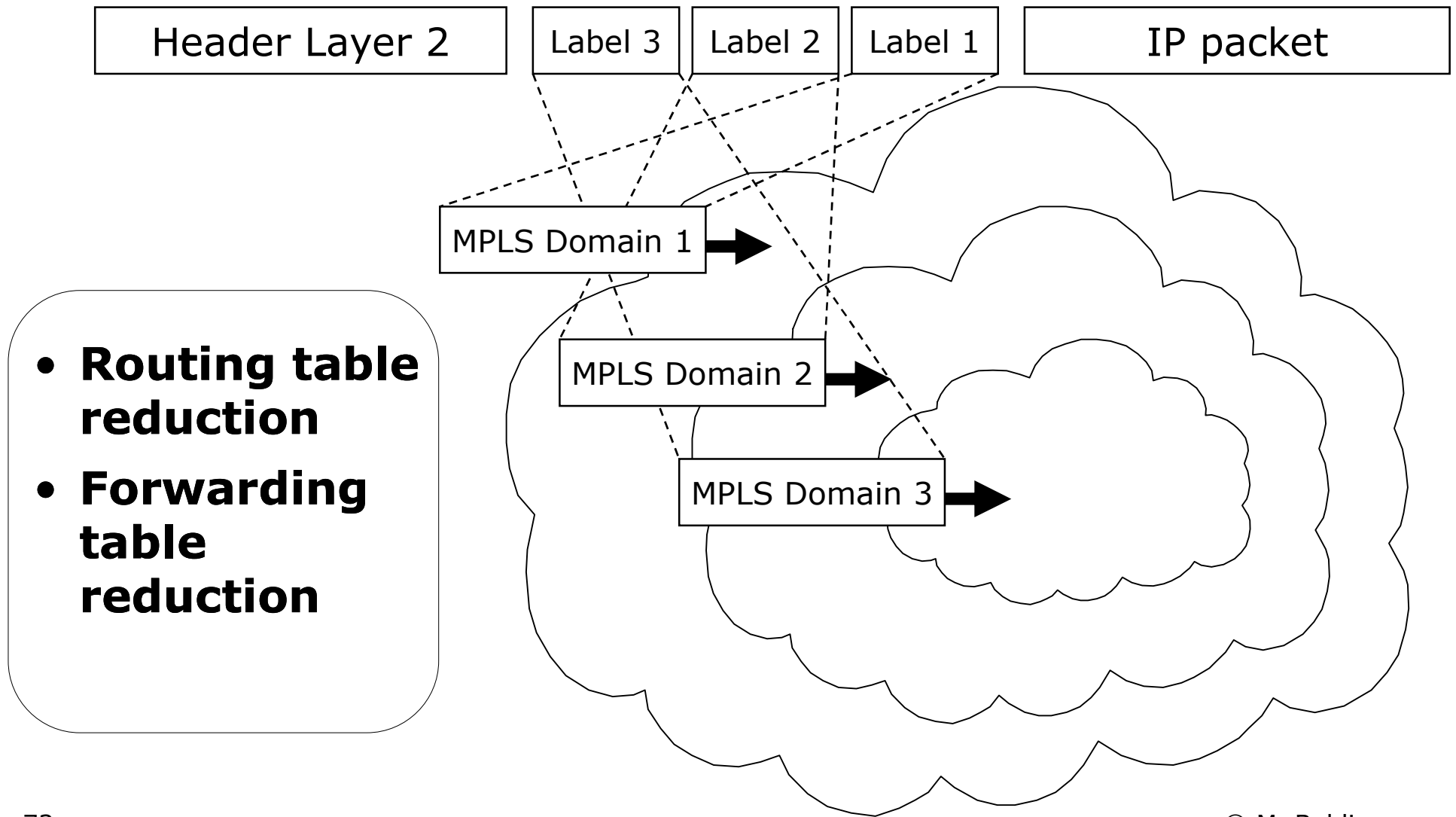


Fast Fault Recovery

Edge-to-edge re-routing



Label Stack hierarchy and scalability



MPLS and scalability

- MPLS labels introduce hierarchy**
 - Various hierarchical levels, as needed for the necessary scalability**
- Routing tables of transit routers do not have to be comprehensive**
 - LSP between edge routers**
- (Simpler and faster exact match of labels rather than longest prefix matching)**

Penultimate Hop Popping (PHP)

→ The last but one node on the path of an LSP ^{removes} pops the label

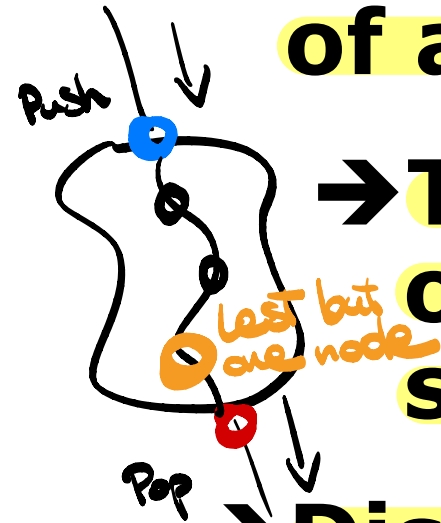
→ The LER routes the packet based on IP address (or next label in stack)

→ Distribution of label 3 indicates (implicit) PHP

→ Explicit PHP: label 0 is swapped

→ When shim header is needed

→ E.g., EXP bits



VPN: Virtual Private Network

Services similar to those of a private network, but provided on a public infrastructure (IP)

- Privacy and security**
- Overlapping addressing spaces (private)**
- Non-unique addresses**
- Class of Service (CoS) and Quality of Service (QoS)**

VPN based on MPLS

- Labels “hide” IP addresses of users on the public network**
- Allow for overlapping address spaces**
- Service flexibility**
 - FEC**

VPN based on MPLS

- MPLS allows a scalable solution for VPN services**
- Other approaches require explicit setting (manual) of tunnels between each pair of websites**
- Number of VPNs**
- Number of VPN members**

Standardization

→ IETF – Internet Engineering Task Force

→ MPLS working group

→ FR/MPLS Alliance

→ Consortium of vendors

→ Foster deployment

→ Aspects omitted by IETF

→ VoMPLS, ADSL

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